GrappStar base station build guide

03 November 2026

Subsystems: drum, frame, launch, reel, sensors. Model exported 2026-11-03 15:57.
All sizes in mm. Masses and print times are estimates.

1. Safety

- Wiring always starts with the battery unplugged. Plug it in last, after a check of every connection.
- Keep hands, hair and loose cloth away from the gears, the reel and the launcher while the motor can run.
- Never stand in front of the launcher. Treat it as loaded whenever the spring is cocked.
- Release the spring before you work on the launcher.
- Wear eye protection when you cut, drill or press parts, and when you test a launch.
- Let hot printed parts and the nozzle cool before you touch them. Print PPA-CF with the enclosure closed.
- Charge LiPo batteries on a fireproof surface and never leave them charging alone.

2. First print: test coupons

Print these before any big part. Each coupon uses the same geometry and numbers as the real part. Files: base/pack/coupons (3MF plates, STL, COUPON_TEST.pdf). Also print the fit coupon (python -m tools.coupon OUT) to tune cad/lib/fits.py.

Print these before the big parts. Each coupon is made from the same geometry functions and numbers as the real part, so a pass here is a pass there. Let parts cool 10 minutes before testing. The fit coupon (tolerance tiles A to H, base/pack/fit_coupon/COUPON.md) is printed alongside.

Settings

PLA: 0.4 mm nozzle, 0.2 mm layers, 4 walls, 40 % gyroid infill unless noted, 5 top and 5 bottom layers, hole and XY compensation 0, no supports. PPA-CF: hardened steel nozzle (never brass), filament dried (80 C, 8 h) and fed from a dry box, 0.2 mm layers, 6 walls, 60 % gyroid infill, 6 top and 6 bottom layers, hole and XY compensation 0, no supports. The 3MF files carry geometry only: set the material and these settings in Bambu Studio.

If a coupon fails

To change a number: on the Mac, after bash kedge/setup_mac.sh (from the repo root), cd base and run ../kedge/.venv/bin/python -m tools.risk_coupon (it rewrites base/pack/coupons and base/pack/fit_coupon). Easier: give the failing result to a Claude session and let it change the numbers. Change one number at a time and reprint only that coupon.

Plates

plate material objects
risk_plate_1_PLA.3mf PLA 24
risk_plate_2_PPA-CF.3mf PPA-CF 26

Tests

1. Swing-arm direction splitter

- Parts: rc1_plate, rc1_sleeve, rc1_sun, rc1_arm, rc1_drag_pad, rc1_drag_shim, rc1_idler_clip, rc1_idler, rc1_output, rc1_crank, rc1_stop, rc1_stop_e
- Bought: 4 x M3x16 (two stop fingers); everything else printed
- Do: Push the drag pad root block and the shim into the arm hub pocket (finger in its recess). On the plate: sleeve on the centre pin, sun on its rib, arm on the sleeve above the sun with its pin down through the idler, printed C clip into the pin groove, output on its pin, crank on top, the two stop fingers on their columns (2 x M3x16 each). Turn the crank 10 turns each way.
- Pass: Clockwise seen from above: the arm swings the idler into the output at once and comes to rest on the engaged stop finger, the output turns, no skipped teeth. Brake the output lightly with a finger and keep turning: the arm stays on that finger (the idler is held at full depth, never pulled in deeper) and the crank still turns smoothly, no binding. Anticlockwise: the arm swings out to the other finger within about 20 deg of crank, the idler runs free and the output stays still. No jam at engagement. The pad drags the arm reliably both ways and the clip stays on.
- If it fails, change: launch_aim.py: FIT (pin and bore play), DRAG_INT and DRAG_SHIM_ADJ (drag: or print a thicker shim), clips.py for the C clip, A() alpha_d = alpha_e + 20 (swing out), _stop_post_xy (swung-out stop), _stop_post_e_xy (engaged stop: binding with the output braked means it lets the idler in too deep); launch_gears.py center_distance extra / spur backlash (mesh).

- Made from: launch_aim.sun_solid, arm_solid, idler_solid (m1.5 z30/18/30, face 12, FIT 0.35), launch_gears.spur for the output, launch_aim.A: centre distances 36.10 / 36.10, output at (11.7, 68.4), engaged arm 96.2 deg on the engaged stop (launch_aim._stop_post_e_xy), swung out 116.2 deg on the other stop (launch_aim._stop_post_xy), sleeve rib launch.RIB_W 5.8 / RIB_R 12.6; drag_pad_solid, drag_shim_solid (DRAG_INT 0.45, shim up to +0.25), pin_clip_solid

2. Printed contact switch

- Parts: rc2_switch_base, rc2_switch_lever

- Bought: M3x10 SHCS ISO 4762 + M3 jam nut, M3 nut, 2x M3x10 self tapping (pan or SHCS) through the lever root into the pad, at most M3x11.7, 2 wires, multimeter

- Do: Nut into the pillar trap (drop of glue), lever clamped with 2 M3 self tappers, contact screw head down through the lever, set the head 0.9 mm above the nut (0.5 mm per turn), lock the jam nut. Meter on beep between screw and nut. Press the lever tip 100 times.

- Pass: Measure the tip travel with calipers (depth rod from the lever tip down to the base top, note the reading at rest, press until it beeps, read again): beep within 0.8 to 1.4 mm of tip travel (design 1.14), silent when let go, tip stops on the hard stop without cracking, lever back flat, same after 100 presses.

- If it fails, change: cad/lib/switch.py contact_switch args travel, overtravel, t, b; STRAIN_LIMIT, NUT_PRESS (loose nut), SCREW_LENGTHS.

- Made from: cad.lib.switch.contact_switch(lever_len=36) with the library defaults (travel 0.8, overtravel 0.6, beam t 1.6): gap 0.91, pretravel at the tip 1.14, close force 0.83 N, strain 0.0061

3. Dog clutch and compliant fork

- Parts: rc3_base, rc3_reel_face, rc3_collar, rc3_fork, rc3_dog_post, rc3_rail

- Bought: M3 nut (yoke tab), 2 x M3x12 (SW_DOG screws in the post), 1 x M3 (post clamp), multimeter

- Do: Press the rail into the base. REEL face on the journal (dogs up), collar on the hex (dogs down), fork pads in the collar groove and its sleeve on the rail, dog post clamped on the rail below the fork, SW_DOG screws in the post set 0.5 mm short of the tab nut at full engagement. Push the fork 7 mm down with the dogs tip to tip, then turn the REEL face until they drop in. Turn the face hard both ways, then pull the fork back. Meter on beep between the lower screw and the tab nut.

- Pass: Yoke flexes when tip to tip (no jam, roofs slide off), dogs drop in within a quarter turn, under hand torque the dogs pull in, never walk out, a light pull on the fork releases them, the yoke springs back flat; the meter beeps only with the dogs fully in, never tip to tip.

- If it fails, change: _drum_kin.py DEFAULTS dog_backcut, dog_roof, dog_deg, dog_gap, fork_k; drum.py GEOM fork_t, yoke_r, groove, dog_touch, dogpost_s; fits.py SHAFT_IN_BORE (collar on the hex, face on the journal), HOLE_SLIDE (fork on the rail).

- Made from: drum.dog_teeth (dog_n 4, dog_h 5, dog_deg 36, backcut 4, roof 1.2, r 18..26), drum.fork, drum.dog_post (dog_touch 0.5), drum.rail, drum.hexprism, GEOM journal d33, groove (82.0, 88.0), stroke 7, neutral gap 2.5

4. Face slip clutch with the flexure spring

- Parts: rc4_hub_plate, rc4_friction_disc, rc4_pressure_plate, rc4_flexure, rc4_adjuster_plate

- Bought: 3 x M3x20 SHCS, 3 x M3 nut, string, coins in a small bag, kitchen scale

- Do: Stack on the key stub: hub plate (counterbores down), friction disc, pressure plate (pads up), flexure (beam down, tabs on the screws), adjuster plate (traps up); 3 screws from below, nuts on top. Run the nuts down until the flexure just touches (zero), then turn all 3 nuts the same, 1/4 turn (0.125 mm) at a time. Clamp the stub level in a vice, tie a string to a disc lug (r 28.5 mm), lay it over the rim and hang a bag of coins weighed on a kitchen scale. Add or take coins to find the hold and slip weights at each setting.

- Pass: At each setting the lower weight must not pull the disc round and the higher one must (smooth steady slip, no judder); same result 3 times; the flexure never bottoms before 1 turn. Bands: 1/4 turn: 35 g holds, 60 g slips (design 48 g, 0.013 N m); 1/2 turn: 70 g holds, 120 g slips (design 95 g, 0.027 N m); 3/4 turn: 105 g holds, 180 g slips (design 143 g, 0.040 N m); 1 turn: 140 g holds, 240 g slips (design 190 g, 0.053 N m).

- If it fails, change: cad/lib/clutch.py: slip_clutch args flex_t, flex_gap, flex_ro, r_in, r_out, n_grooves; MU_PLA_PLA, E_PLA, SIGMA_FLEX_PLA_SUSTAINED (creep: if the torque falls overnight).

- Made from: cad.lib.clutch.slip_clutch() defaults (reel does not use it: its clutch is the spool face plus the PETG rocker): shaft d20 key 6, friction r 18.5..27, mu 0.3, flexure t 0.8 gap 0.5, 0.106 N m per mm, travel 0.50; on the shared keyed stub of coupon 7

5. Bearing seats

- Parts: rc5_seats_608_693, rc5_csk20_seat

- Bought: 608ZZ, F693ZZ, 693ZZ, CSK20PP

- Do: Press each bearing in square with a flat block or a vice (tile left to right: 608, F693ZZ, 693ZZ); CSK20PP into the round boss with its outer keyway on the printed key.

- Pass: Firm press, seated flat on the lip, stays in when tapped upside down, spins as freely as before; the CSK20PP key slides in without force, it still freewheels one way and locks the other.

- If it fails, change: fits.py BRG_RIB_INTERF, BRG_RIB_BORE (all seats); bearings.py CSK20_KEY_RCLEAR, CSK20_OKEY_W (key), KEY_CLEAR in fits.py. If the 608 is off, print tile A of the fit coupon (base/pack/fit_coupon, COUPON.md).

- Made from: cad.lib.bearings.seat_608(bottom lip, lip_d 14) as reel, seat_f693(), seat_693(), seat_csk20(key_deg 90) as the reel capstan block (wall 4); fits BRG_RIB_INTERF 0.15, BRG_RIB_BORE 0.3, CSK20_KEY_RCLEAR 0.2

6. M3 holes

- Parts: rc6_m3_tile

- Bought: M3x4x5 heat-set insert, M3 nut, 2 x M3x10

- Do: Left to right: heat-set the insert (220 C), press the nut in, push a screw through, self-tap a screw in.

- Pass: Insert flush and does not spin when the screw is tightened; nut presses in and holds; screw drops through the clearance hole; the tap hole cuts a thread and holds a hard pull without splitting.

- If it fails, change: fits.py M3_INSERT_D, M3_NUT_AF, M3_CLEAR_D, M3_TAP_D (tiles C, D, E of the fit coupon in base/pack/fit_coupon bracket them).

- Made from: cad.lib.fasteners.m3_insert (d 4.0), m3_nut_trap (AF 5.8), m3_clear (d 3.4), m3_tap (d 2.6)

7. Key rib and spigot

- Parts: rc7_key_stub, rc7_key_bore, rc7_spigot, rc7_bushing

- Bought: CSK20PP from coupon 5 (inner keyway)

- Do: Slide the key stub into the keyed tile and into the CSK20PP inner race (the real interface). Push the spigot into the sleeve slice and turn it.

- Pass: Stub slides in by hand, key in the keyway, no visible rock. Play: clamp the tile, tape a ruler to the stub end, mark the ruler 60 mm from the axis, rock the stub both ways: the mark moves under 1 mm (about 1 deg). The spigot turns freely; rocked at its free end it moves under 0.5 mm.

- If it fails, change: fits.py KEY_CLEAR, SHAFT_IN_BORE; shafts.py KEY20_H, SHAFT_PRINT_UNDERSIZE; launch.py SPIGOT_BORE (now 12.35); drum.py GEOM spig_d.

- Made from: cad.lib.shafts.keyed_shaft(d 20, key_w 6, key_h 2.6) and keyed_bore (clear 0.3); drum.main_shaft (spig_d 12); launch.sleeve_solid (SPIGOT_BORE 12.35)

8. Gear meshes m1.25, m1.5, m2, m2.5

- Parts: rc8_m125_plate, rc8_m125_seg, rc8_m125_pinion, rc8_m15_plate, rc8_m15_z42, rc8_m15_z24, rc8_m2_plate, rc8_m2_z50, rc8_m2_z22, rc8_m25_plate, rc8_m25_big, rc8_m25_drive

- Bought: none

- Do: Each pair on its plate (label = module), flat faces down; turn or rock through every tooth.

- Pass: Smooth, no tight spot, a little play when rocked; m2.5: the tall tooth and the cut-down last tooth pass without catching, then the mesh lets go at the gap.

- If it fails, change: fits.py GEAR_BACKLASH, GEAR_CENTER_EXTRA (m1.25, m1.5, m2); launch_gears.py center_distance extra, spur backlash, sector_gear_2d tall_factor (m2.5); _drum_kin.py pin_x (m1.25 pinion).

- Made from: m1.25: gears.spur_gear z60 sector / z10 shift 0.4 tip_short 0.08 (drum.tip_short_for), cd 44.33 (drum.a_seg); m1.5: gears.spur_gear z42 sector / z24, cd 49.60; m2: gears.spur_gear z50 sector (drum GEOM out_m, out_z) / z22 (reel Z_CAPG), cd 72.10; m2.5: launch_gears.sector_gear_2d z40 (n_present 27, last 6 + gap) / launch_gears.spur z26, cd 82.60 (launch L); coupon face 6, fits GEAR_BACKLASH 0.15, GEAR_CENTER_EXTRA 0.1

9. Shear-pin torque limiter

- Parts: rc9_hub_lever, rc9_outer_lever, rc9_shear_pin x3

- Bought: string or a strap, luggage scale

- Do: Print the pins lying down on their flat, as exported (D44: shear plane across the layers; never stand them up). Hub slice into the outer slice, pin through both holes (head outside). Clamp the hub lever, hang the luggage scale on the outer lever hole (r 60 mm) at right angles and pull slowly until the pin shears. Break all 3 pins. Torque = kg x 9.81 x 0.06 N m.

- Pass: Each pin shears cleanly between 5.9 and 10.2 kg on the scale (3.5 to 6 N m, drum_NOTES; design 4.9 N m = 8.3 kg), the 3 pins within 20 percent of each other, both slices undamaged (holes still round); a fresh pin goes in by hand.

- If it fails, change: drum.py GEOM shear_d (torque goes with d squared), hub_od (radius), shear_s; fits for the hole: drum.coupler shear hole (shear_d + 0.1) and shear_pin (shear_d - 0.04). Spec D8 / D26 limit 5 N m.

- Made from: drum.coupler_hub and drum.coupler (sliced s 4.5..20), drum.shear_pin: d4 PLA in single shear at r 13 (shear_s 9); 4.9 N m at 30 MPa

10. AS5600 magnet cap (bench test)

- Parts: rc10_magnet_cap

- Bought: magnet: not included with the board, see MEASURE_LIST C13 (any small neodymium disc edge-on, or 6 x 2.5 diametric), M3x6, superglue

- Do: Glue the magnet into the end pocket (flush, any way round), push the cap on the shaft end (flat to the flat), M3x6 onto the flat. It fits the d12 main shaft spigot of the base (for the bare-motor bench test, tape the magnet to the wiper shaft end instead). AS5600 chip on the shaft axis, 1 to 2 mm from the magnet face.

- Pass: Cap does not wobble or slip on the shaft; the AS5600 angle follows the shaft smoothly through full turns (no jumps), magnet status OK (not too weak, not too strong) at 1 to 2 mm.

- If it fails, change: drum.py GEOM cap_s, spig_d; magnet_holder pocket d6.2 x 2.6 (magnet loose or tight); fits.py HOLE_SLIDE (cap on the spigot).

- Made from: drum.magnet_holder (spig_d 12 with its flat, cap_s (195.0, 201.5), magnet pocket d6.2 x 2.6), printed as drum prints it

Parts

part material qty print bbox mm what
rc1_plate PLA 1 110.1 x 82.2 x 33.1 Splitter base: centre pin d8, output pin d8, two stop finger columns (2 x M3 tap each)
rc1_sleeve PPA-CF 1 26.0 x 26.0 x 29.3 Sleeve stub d20 with the integral key rib (launch.RIB_W, RIB_R) under the sun and the crank
rc1_sun PPA-CF 1 47.8 x 48.0 x 12.0 launch_aim.sun_solid: sun m1.5 z30 as built
rc1_arm PPA-CF 1 61.4 x 41.0 x 19.3 launch_aim.arm_solid: swing arm as built (drag pad pocket, d8 pin, clip groove)
rc1_drag_pad PLA 1 24.6 x 8.7 x 4.0 launch_aim.drag_pad_solid: printed flexure drag pad as built
rc1_drag_shim PLA 1 9.4 x 3.0 x 4.0 launch_aim.drag_shim_solid: nominal preload shim as built
rc1_idler_clip PLA 1 10.8 x 10.0 x 1.5 launch_aim.pin_clip_solid: printed C clip for the d8 idler pin
rc1_idler PPA-CF 1 30.0 x 29.7 x 12.0 launch_aim.idler_solid: idler m1.5 z18 as built
rc1_output PPA-CF 1 47.8 x 48.0 x 12.0 Output m1.5 z30 (launch_gears.spur, face 12) on a d8 pin
rc1_crank PPA-CF 1 48.2 x 30.3 x 20.0 Hand crank, keyed on the sleeve rib like the sun
rc1_stop PLA 1 47.1 x 10.0 x 6.4 Stop finger: rounded end at the as-built swung-out stop post position
rc1_stop_e PLA 1 64.4 x 10.0 x 6.4 Engaged stop finger: rounded end at the as-built engaged stop post position (launch_aim._stop_post_e_xy)
rc2_switch_base PLA 1 54.0 x 35.0 x 11.5 contact_switch base: pad, nut pillar, hard stop, wire channel
rc2_switch_lever PLA 1 50.0 x 5.0 x 8.0 contact_switch lever, printed on its side (Y up)
rc3_base PLA 1 40.0 x 91.5 x 97.5 Base: d30 journal (REEL sleeve) and hex 22 AF (main shaft), rail hole (press)
rc3_reel_face PPA-CF 1 61.6 x 54.0 x 9.5 REEL gear dog face (drum.dog_teeth, phase 45) with a handle
rc3_collar PPA-CF 1 60.0 x 60.0 x 18.0 Collar lower half: drum.dog_teeth facing down, fork groove, hex bore

rc3_fork PLA 1 87.1 x 94.1 x 83.0 drum.fork as built: compliant yoke, rail sleeve, follower pin, M3 nut in the tab
rc3_dog_post PLA 1 20.0 x 75.9 x 41.7 drum.dog_post as built: SW_DOG_L / SW_DOG_R screw bosses, clamps on the rail
rc3_rail PPA-CF 1 110.0 x 10.0 x 9.0 drum.rail profile d10 with the print flat, cut to 110 long
rc4_hub_plate PLA 1 54.0 x 54.0 x 6.0 clutch.slip_clutch hub_plate (library defaults)
rc4_friction_disc PLA 1 57.0 x 55.0 x 3.0 clutch.slip_clutch friction_disc (library defaults)
rc4_pressure_plate PLA 1 54.0 x 54.0 x 3.5 clutch.slip_clutch pressure_plate (library defaults)
rc4_flexure PLA 1 50.0 x 50.0 x 1.7 clutch.slip_clutch flexure (library defaults)
rc4_adjuster_plate PLA 1 54.0 x 54.0 x 4.0 clutch.slip_clutch adjuster_plate (library defaults)
rc5_seats_608_693 PLA 1 60.0 x 28.0 x 9.6 608 (reel spool seat), F693ZZ and 693ZZ crush-rib seats
rc5_csk20_seat PLA 1 55.3 x 55.3 x 16.0 CSK20PP seat with the printed key (reel capstan block, key_deg 90)
rc6_m3_tile PLA 1 54.0 x 18.0 x 8.6 M3 heat-set hole, nut trap, clearance and tap holes (fits.py values)
rc7_key_stub PPA-CF 1 21.2 x 20.0 x 31.0 shafts.keyed_shaft d20 key 6 (reel capstan tube numbers), 31 long, also the coupon 4 clutch shaft
rc7_key_bore PPA-CF 1 36.0 x 36.0 x 8.6 shafts.keyed_bore d20 key 6, 8 long (keyed hubs on the sleeve rib are PPA-CF)
rc7_spigot PPA-CF 1 28.0 x 25.4 x 22.0 drum.main_shaft slice: hex 8 long + d12 spigot 20 long, on a hex flat
rc7_bushing PPA-CF 1 32.0 x 32.0 x 16.0 launch.sleeve_solid slice 16 long: spigot bore d12.35
rc8_m125_plate PPA-CF 1 62.3 x 18.0 x 11.0 Pins at 44.33
rc8_m125_seg PLA 1 38.1 x 46.7 x 6.0 drum segment gear m1.25 z60 sector
rc8_m125_pinion PPA-CF 1 15.8 x 15.2 x 6.0 drum wrap pinion m1.25 z10 x0.4 tip_short 0.08
rc8_m15_plate PPA-CF 1 67.6 x 18.0 x 11.0 Pins at 49.60
rc8_m15_z42 PPA-CF 1 41.0 x 29.6 x 6.0 drum wrap idler output m1.5 z42 sector
rc8_m15_z24 PPA-CF 1 38.9 x 38.9 x 6.0 reel cup ring gear m1.5 z24
rc8_m2_plate PPA-CF 1 90.1 x 18.0 x 11.0 Pins at 72.10
rc8_m2_z50 PPA-CF 1 61.0 x 39.3 x 6.0 drum REEL output m2 z50 sector
rc8_m2_z22 PPA-CF 1 48.0 x 47.7 x 6.0 reel capstan gear m2 z22
rc8_m25_plate PPA-CF 1 100.6 x 18.0 x 11.0 Pins at 82.60
rc8_m25_big PPA-CF 1 61.5 x 49.0 x 6.0 launch big gear m2.5 z40 sector: teeth 21..26 (tall, relieved)
rc8_m25_drive PPA-CF 1 42.5 x 44.0 x 6.0 launch drive gear m2.5 z26 sector
rc9_hub_lever PPA-CF 1 79.0 x 26.0 x 15.5 drum.coupler_hub slice (shear pin hole) with a lever, string hole at r 60
rc9_outer_lever PPA-CF 1 86.0 x 40.0 x 9.5 drum.coupler slice (shear pin hole) with a lever, string hole at r 60
rc9_shear_pin PLA 3 12.1 x 5.0 x 4.1 drum.shear_pin as built (d4 PLA), printed LYING on its 0.4 mm flat as drum prints it (D44): the shear plane runs across the layers
rc10_magnet_cap PLA 1 20.0 x 20.0 x 6.5 drum.magnet_holder as built: AS5600 magnet cap on the d12 spigot end

3. Parts list

Printed parts

part	subsystem	material	qty	mass g	notes
drum_bracket_a	drum	PLA	1	37.7	Drum bracket A: axle, shift rail, N20 ear, 2 x M3 to the deck
drum_bracket_b	drum	PLA	1	23.4	Drum bracket B: axle, shift rail, 2 x M3 to the deck
drum_bracket_idler	drum	PLA	1	12.5	Wrap idler journal post, 2 x M3 to the deck
drum_bracket_idler2	drum	PLA	1	16.2	Wrap idler second journal post (motor end), 2 x M3 to the deck
drum_dog_post	drum	PLA	1	6.0	Dog contact post on the shift rail: SW_DOG_R and SW_DOG_L screws (collar fully engaged)
drum_drum_a	drum	PLA	1	146.5	Drum A: N20 gear 120T m0.8, brake cam, wrap segment gear + locking arc, hex spigot
drum_drum_b	drum	PLA	1	79.0	Drum B: shift groove barrel, Gray code bands A B C, hand lever hex boss
drum_fork	drum	PLA	1	25.9	Compliant shift fork: thin yoke is the spring (about 1.5 N/mm), follower pin in groove 1, contact tab
drum_hand_lever	drum	PLA	1	4.0	Hand lever on the drum hex boss (fallback: swing the N20 out first)
drum_index_block	drum	PLA	1	8.1	SW_INDEX block: 2 x M3 x 12 through the idler post 2 strut, M3 nut contact
drum_index_lever	drum	PLA	1	0.9	SW_INDEX lever on the 6-bump coupler cam (D12 fallback encoder)
drum_magnet_cap	drum	PLA	1	1.9	AS5600 magnet cap on the spigot end (6 x 2.5 diametric or edge-on disc), M3 x 6 onto the flat

part	subsystem	material	qty	mass g	notes
drum_n20_cradle	drum	PLA	1	11.9	N20 cradle on the bracket A ear: pivot + swing screw (hand fallback), SW_N20 contact
drum_n20_strap	drum	PLA	1	2.7	End plate closing the N20 cradle, 2 x M3 x 8
drum_shear_pin	drum	PLA	6	0.2	Torque limiter shear pin d4 (about 5 N m), prints lying; print 6, 5 are spares
drum_sw_lever_0	drum	PLA	1	1.0	Code switch lever A: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_1	drum	PLA	1	1.0	Code switch lever B: flexure, round nub, M3 x 6 contact screw
drum_sw_lever_2	drum	PLA	1	1.0	Code switch lever C: flexure, round nub, M3 x 6 contact screw
drum_switch_block	drum	PLA	1	16.6	Code switch block: 3 lever feet, 3 M3 nut contacts (SW_DRUM_A, B, C)
frame_cable_anchor_1	frame	PLA	1	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_cable_anchor_2	frame	PLA	1	4.0	cable-tie anchor (strain relief), 1 x M3 x 12 into the deck
frame_camera_mount	frame	PLA	1	166.5	Kiyo X mount: its own clip grips the 16 mm edge plate (or a 1/4-20 screw), 4 x M3 x 10 into the deck
frame_can_stay	frame	PLA	1	28.4	can stay: post on the motor plate (2 x M3 x 12 into plate inserts), cable tie or hose clamp round the can through the cradle slots; reprint with the adapter
frame_cheek_r	frame	PLA	1	182.5	motor plate (y -10..0, front face = the spec's y 0): window for the puck adapter, 8 rim holes (M3 x 12 from the front into the puck lip), bridge to the motor leg (2 x M3 x 12), can stay and hood inserts; carved round the other subsystems
frame_deck_1	frame	PLA	1	291.2	deck: M3 inserts for every frame foot; 9 x M3 rover holes (it is its own drill template)
frame_deck_2	frame	PLA	1	241.7	deck: M3 inserts for every frame foot; 7 x M3 rover holes (it is its own drill template)
frame_deck_3	frame	PLA	1	300.1	deck: M3 inserts for every frame foot; 6 x M3 rover holes (it is its own drill template)
frame_deck_4	frame	PLA	1	248.7	deck: M3 inserts for every frame foot; 10 x M3 rover holes (it is its own drill template)
frame_ebay	frame	PLA	1	175.3	electronics bay: ESP32-S3 cradle (USB slot), BTS7960 on standoffs with vents, DRV8833, MP1584, HX711 pockets, fuse cradle, wire exits with tie anchors
frame_ebay_esp_clamp	frame	PLA	1	1.6	ESP32 hold-down over the module end, 2 x M3 x 8 into the cradle posts
frame_ebay_lid	frame	PLA	1	71.4	bay lid: grille over the BTS7960, 4 x M3 into the corner bosses
frame_guard_front	frame	PLA	1	37.1	front gear guard between the drum gears and the bay, M3 x 10 into the deck
frame_guard_hood	frame	PLA	1	26.2	gear hood over the reel and bus gears, M3 x 10 through its lugs into the motor plate
frame_motor_leg	frame	PLA	1	117.3	motor support leg (prints upright): 4 x M3 x 10 into the deck, 2 x M3 x 12 along y into the bridge behind the motor plate
frame_rc832_holder	frame	PLA	1	22.8	RC832 holder: receiver stands antenna-end up in the slot, leads out low, 2 x M3 x 10
launch_aim_cap	launch	PLA	1	5.2	Worm shaft saddle cap, 4 x M3x30 through the legs into 13 deep pilots in the saddles
launch_aim_crankpin_clip	launch	PLA	1	0.1	Printed C clip on the crank pin, inboard of the conrod

part	subsystem	material	qty	mass g	notes
launch_aim_drag_pad	launch	PLA	1	0.4	Flexure drag pad: cantilever finger t 2, tip bump 0.45 into the sleeve d20 (strain 0.43 %, about 2.3 N), root block pushed axially into the arm hub pocket
launch_aim_drag_shim	launch	PLA	1	0.1	Preload shim behind the drag pad root block (nominal; print up to 0.25 thicker for more drag, strain stays under 0.7 %)
launch_aim_housing	launch	PLA	1	185.4	Aim housing on the deck (4 x M3 x 10 from the top through its counterbored foot flanges into the deck inserts): sleeve end bushing, countershaft bushings, worm saddles, M8 wheel bolt (head on the outer boss, nut in the inner face hex pocket), engaged and swung-out swing arm stop posts
launch_aim_idler_clip	launch	PLA	1	0.1	Printed C clip on the arm pin, holds the idler
launch_aim_lugpin_clip	launch	PLA	1	0.1	Printed C clip on the lug pin, outboard of the conrod
launch_carriage	launch	PLA	1	59.6	Carriage: rack, shoes, dart cradle, pusher with tether hole, spring arms and pegs, SW feet
launch_carriage_yoke	launch	PLA	1	23.5	Bolted yoke: both spring arms (hit the stop, d8.1 holes for the 8 mm steel rod spring pegs, D5 holes at y +-56 empty), cross bar, keys, SW_SEATED block, rigid SW_HOME foot. Prints on the peg boss bottoms and key plates (w 62)
launch_drive_washer	launch	PLA	1	0.2	Spacer washer: drive gear hub to the left cheek
launch_gear_clip	launch	PLA	1	0.3	C clip on the big gear pin groove
launch_gear_washer	launch	PLA	1	0.5	Spacer washer: big gear hub to the left cheek
launch_guard_l_a	launch	PLA	1	65.0	Spring guard left rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b1	launch	PLA	1	4.5	Spring guard left mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_l_b2	launch	PLA	1	17.9	Spring guard left mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_l_c	launch	PLA	1	4.8	Spring guard left front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_guard_r_a	launch	PLA	1	65.0	Spring guard right rear (end wall, top cover) (D33): 2 mm side plate outside both spring pairs, 4 wall brackets (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b1	launch	PLA	1	4.5	Spring guard right mid front of the stop frame (D33): 2 mm side plate outside both spring pairs, 1 wall bracket (M3x12 self tap into the rail wall); fit by moving it inward
launch_guard_r_b2	launch	PLA	1	21.8	Spring guard right mid under the stop spring (D33): 2 mm side plate outside both spring pairs, 2 wall brackets (M3x12 self tap into the rail wall) and a foot on the front plate (M3x10 self tap down); fit by moving it inward
launch_guard_r_c	launch	PLA	1	4.8	Spring guard right front (top cover) (D33): 2 mm side plate outside both spring pairs, 2 bosses on the front plate edge (M3x12 self tap along y); fit by moving it inward
launch_muzzle_roller	launch	PLA	1	0.5	Lip roller at the carriage front: the tether bends down over it in tow and retrieval (groove r 6 for the 1 mm line), 693ZZ inside

part	subsystem	material	qty	mass g	notes
launch_muzzle_washer_l	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_muzzle_washer_r	launch	PLA	1	0.0	Lip roller thrust washer (0.9 thick, flange on the cheek side) between the bearing inner race and the cheek: put both in the roller bore, then drop roller and washers into the fork slot
launch_pulley_bot	launch	PLA	1	0.5	Tether pulley (bot), groove r 6, 693ZZ inside
launch_pulley_bot_washer_l	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_pulley_bot_washer_r	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_pulley_piv	launch	PLA	1	0.5	Tether pulley (piv), groove r 6, 693ZZ inside
launch_pulley_top	launch	PLA	1	0.5	Tether pulley (top), groove r 6, 693ZZ inside
launch_pulley_top_washer_l	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_pulley_top_washer_r	launch	PLA	1	0.1	Printed top-hat thrust washer between fork plate and 693ZZ inner race
launch_rail_front	launch	PLA	1	137.7	Rail front piece: shoe track end, spring anchor posts with straps, muzzle tie
launch_rail_mid	launch	PLA	1	183.0	Rail mid piece: shoe track, cheeks on the sleeve, big gear pin bosses, lever seats, SW_GAP bracket pilots
launch_rail_rear	launch	PLA	1	135.7	Rail rear piece: shoe track, walls, tether fork and two pulleys behind the rear end
launch_rail_splice_l	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_rail_splice_r	launch	PLA	1	0.3	Top splice strap over the front rail joint (4 x M3 x 10 self tap)
launch_seat_clip	launch	PLA	1	0.1	Head cap on the plunger pin (rests on the yoke block)
launch_stop_frame_l	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_frame_r	launch	PLA	1	32.7	Stop frame (print outboard face down): outboard preload ledge, anchor seat, rear foot (2 x M3x12 into the rail wall), front foot (2 x M3x16 into the rail front plate, hex key through the d6.5 holes in the front wall). Fit before the spring guards and the main springs
launch_stop_spring_l	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_stop_spring_r	launch	PLA	1	81.6	Carriage stop flexure: 16 loops (31 tapered leaves t 3.15/1.0 x 14), print upside down on the flat w 83.5 face, 100% infill / all perimeters; preload 6.6 mm by 2 x M3x20 draw screws through the frame front wall
launch_sw_home	launch	PLA	1	0.8	SW_HOME printed flex lever on the right rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)

part	subsystem	material	qty	mass g	notes
launch_sw_seated	launch	PLA	1	0.8	SW_SEATED printed flex lever on the left rail lip, at rest (prints on its side): beam 0.8 thick, contact gap 0.50 (one screw turn), pad pretravel 0.70 + overtravel 0.25, peak strain 0.10 % (limit 0.8 %)
reel_bar_cap_1	reel	PLA	1	1.0	Bar guide cap: closes the U guide over the bar. Print flat
reel_bar_cap_2	reel	PLA	1	1.0	Bar guide cap: closes the U guide over the bar. Print flat
reel_c_clip	reel	PLA	1	1.0	C-clip holding the capstan gear on the tube
reel_cup	reel	PLA	1	12.7	Wrap cup (Harry v7): ring and groove lay the wraps on the capstan. Print open end down
reel_cup_clip	reel	PLA	1	0.3	Harry's v7 C-clip on the cup shaft neck in front of the cup
reel_k_clip	reel	PLA	1	0.1	C-clip holding the cup gear on the cup shaft
reel_p2_clip	reel	PLA	1	0.2	C-clip on the swivel tube above the post block
reel_pulley1	reel	PLA	1	1.2	Pulley 1: turns the line from the exit eye toward the rail (D14)
reel_pulley2	reel	PLA	1	1.2	Pulley 2: feeds the line to cad-launch's rail pivot pulley (D34)
reel_s_clip	reel	PLA	1	0.1	C-clip holding the reel gear
reel_shroud	reel	PLA	1	14.5	Shroud round the storage reel: keeps loose loops on it. Print standing on its flange
reel_spool	reel	PLA	1	34.3	Storage reel body: 20 m of 1 mm Dyneema on a dia 50 core, two 608s. Print rear flange down
reel_spool_flange	reel	PLA	1	9.9	Front flange disc: keeps the front 608 in, carries the clutch friction ring. Print flat
reel_sw_base	reel	PLA	1	7.0	SW_ARM contact base (printed switch, M3 head on M3 nut)
reel_sw_bracket	reel	PLA	1	20.6	SW_ARM bracket: screwed to the wall below the gears. Print flat
reel_sw_lever	reel	PLA	1	1.3	SW_ARM lever: rides the cup gear's raised track (contact closed) and drops into its home notch (open). Print on its side
reel_t	reel	PLA	1	1.7	Thrust ring: the rocker pushes it onto the rear 608 inner race
reel_wall	reel	PLA	1	204.8	Reel wall: foot, guide eye post, rocker ears, stop pins, brake guides and crank boss. Print on its foot
sensors_as5600_carrier	sensors	PLA	1	12.7	AS5600 carrier: plate with 4 pegs through the board holes, column screwed to the aim housing spine
sensors_as5600_strip_bot	sensors	PLA	1	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_as5600_strip_top	sensors	PLA	1	1.0	AS5600 clamp strip on the board edge over the pegs, M3 from the strip front into a nut in the carrier back
sensors_sw_gap	sensors	PLA	1	9.9	SW_GAP printed contact base (cad.lib.switch, coupon 2 geometry): with its arm screwed to the right rail cheek outer face below the big gear, closed by the big gear cam bump (gear in its gap)
sensors_sw_gap_lever	sensors	PLA	1	1.4	SW_GAP lever, t 1.4 mm, print on its side so layers run along the beam
sensors_sw_rail_low	sensors	PLA	1	7.3	SW_RAIL_LOW printed contact base (cad.lib.switch, coupon 2 geometry): on its bracket on the aim housing foot, closed by the crank cam at the LOW dead point

part	subsystem	material	qty	mass g	notes
sensors_sw_rail_low_bracket	sensors	PLA	1	17.8	SW_RAIL_LOW bracket on the aim housing foot extension: an M3x10 into the extension and its own deck screw, an M3x16 from the top through the counterbored foot and the extension into a deck insert
sensors_sw_rail_low_lever	sensors	PLA	1	1.3	SW_RAIL_LOW lever, t 1.4 mm, print on its side so layers run along the beam
drum_axle	drum	PPA-CF	1	13.1	Fixed drum axle d10, pressed into brackets A and B
drum_collar	drum	PPA-CF	1	44.0	Dog collar: hex bore, 4 back-cut roofed dogs each face, fork groove (bridges over the lower dogs)
drum_coupler	drum	PPA-CF	1	89.3	Coupler: 5 N m shear pin limiter (D8), bus gear 50T m1.5 (D25), SW_INDEX cam, REEL journal, hex socket; prints hub end down, tree support under the bus gear ring
drum_coupler_hub	drum	PPA-CF	1	14.6	Coupler hub on the wiper shaft (bore parametric: WIPER), M8 nut inside, shear pin hole
drum_dogring_launch	drum	PPA-CF	1	19.0	LAUNCH dog ring: 3 slots for the launch sleeve lugs, 3 x M3 from the sleeve side
drum_gear_reel	drum	PPA-CF	1	95.9	REEL output gear m2 50T face 12 (D20), free on its sleeve, dog face toward the collar
drum_idler	drum	PPA-CF	1	7.6	Wrap idler shaft: pinion 10T m1.25 (locked at home by the drum lock arc), lock cam, two journals
drum_idler_gear	drum	PPA-CF	1	27.5	Wrap idler output gear 42T m1.5 (D bore) to the cad-reel cup ring 24T: 7 cup turns
drum_n20_pinion	drum	PPA-CF	1	0.5	N20 pinion 10T m0.8, D bore pressed on the N20 shaft
drum_rail	drum	PPA-CF	1	13.1	Shift rail d10, pressed into brackets A and B
drum_reel_washer	drum	PPA-CF	1	0.9	REEL gear retaining washer on the coupler journal end, 2 x M3 x 6
drum_shaft	drum	PPA-CF	1	45.2	Main shaft: hex 22 AF, d12 spigot into the launch sleeve; prints on a hex flat (support under the spigot)
frame_motor_adapter	frame	PPA-CF	1	276.3	puck adapter: slotted M6 holes (+-4 mm) for the motor bosses, rear lip with 8 M3 inserts for the rim screws, tether channels open to the edge; reprint when the motor is measured
launch_aim_arm	launch	PPA-CF	1	6.6	Swing arm on the sleeve, dragged by the printed PLA flexure pad in its hub; printed idler pin with a printed C clip groove
launch_aim_bevel30	launch	PPA-CF	1	8.8	Bevel m1.5 z30 on the countershaft D flat, drives the z15 2:1
launch_aim_conrod	launch	PPA-CF	1	14.2	Conrod 156 between pin centres (PPA-CF), printed C clips on both pins
launch_aim_crank	launch	PPA-CF	1	35.3	Worm wheel m2 z20 face 16 = crank gear, crank arm r30 and d8 crank pin in one part, two 608 inside
launch_aim_idler	launch	PPA-CF	1	7.5	Splitter idler m1.5 z18 face 12 on the arm pin
launch_aim_lug	launch	PPA-CF	1	16.9	Lug bracket (PPA-CF), 2 x M3 into the rail wall, d8 lug pin at (u 115.2, w 0)
launch_aim_output	launch	PPA-CF	1	26.2	Splitter output m1.5 z30 face 12 with its d8 countershaft (D flat)
launch_aim_sun	launch	PPA-CF	1	18.3	Splitter sun m1.5 z30 face 12, keyed on the launch sleeve
launch_aim_worm	launch	PPA-CF	1	21.3	Worm m2 pd24 single start (lead angle 4.8 deg, self-locking) + d8 shaft + bevel z15
launch_big_gear	launch	PPA-CF	1	177.6	Big gear m2.5: sector layer 27 of 40 teeth (rack, v10 release teeth) + full 40T layer (drive), SW_GAP cam bump, on the rail pin

part	subsystem	material	qty	mass g	notes
launch_drive_gear	launch	PPA-CF	1	29.9	Drive gear m2.5 26T, sprag #1 inside on an integral key rib
launch_gear_pin	launch	PPA-CF	1	5.4	Snap pin d10 for the big gear, C clip; prints lying on its flat
launch_seat_plunger	launch	PPA-CF	1	1.0	SW_SEATED plunger: foot pad + stem with a 45 deg chamfer, clip groove on top
launch_seat_pushrod	launch	PPA-CF	1	0.2	SW_SEATED pushrod d3.2 with the stem slot
launch_sleeve	launch	PPA-CF	1	49.3	Launch sleeve d20: dog ring flange, integral key ribs (sprag #1, splitter sun), journals
launch_yoke_peg_l	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
launch_yoke_peg_r	launch	PPA-CF	1	1.2	Spring peg on the yoke arm (printed PPA-CF lying down, or 8 mm steel rod 20 long): pressed into the blind d8.1 hole from below up to its end (a drop of CA if it is loose), the spring hook sits on the 6 mm below the boss (held at w 60 by the spring line); ends deburred
reel_bar	reel	PPA-CF	1	7.9	Brake bar: cad-drum's track 2 cam pushes its nose, its left end turns the crank. Print flat
reel_bus	reel	PPA-CF	1	26.3	Bus gear m1.5 z46 (free on the capstan tube): driven by the drum's bus gear, drives the swing idler. Print face down
reel_c	reel	PPA-CF	1	27.5	Capstan tube: key rib in the sprag and drum, D flat for the capstan gear. Print standing, front end down
reel_crank	reel	PPA-CF	1	2.1	Brake bell crank: the bar turns it, arm B pushes the rocker leg. Print on its flat side
reel_crank_washer	reel	PPA-CF	1	0.2	Printed washer: keeps the crank on its boss
reel_drum	reel	PPA-CF	1	44.5	Capstan drum dia 40 x 40 (Harry v7), keyed on the tube; PPA-CF: hoop stress 11.6 MPa if the shear pin slips. Print flange down
reel_exit_post	reel	PPA-CF	1	70.6	Exit eye post with the pulley 1 fork: carries the full tow load, so PPA-CF. Print on its foot
reel_gc	reel	PPA-CF	1	18.1	Capstan gear m2 z22: driven by the drum's REEL output gear (50T)
reel_gi	reel	PPA-CF	1	11.3	Swing idler m1.5 z30: meshes the reel gear only on forward wiper turns
reel_gk	reel	PPA-CF	1	9.9	Cup gear m1.5 z24: driven by the drum's wrap idler; the raised track on its back holds SW_ARM closed, the notch at home lets it open
reel_gr	reel	PPA-CF	1	4.7	Reel gear m1.5 z20: turns the clutch disk 2.5 x wiper speed
reel_k	reel	PPA-CF	1	11.8	Cup shaft: runs inside the capstan tube, cup in front, cup gear behind. Print standing
reel_p2_fork	reel	PPA-CF	1	4.0	Pulley 2 swivel fork: turns about the incoming line so the groove follows the line to the rail
reel_p2_post	reel	PPA-CF	1	57.4	Pulley 2 post: stands on the deck beside the drum, turns the line into the rail pivot pulley's plane
reel_rocker	reel	PPA-CF	1	6.1	Brake rocker: flexible PPA-CF lever = clutch spring, pushed by the crank. Print flat
reel_s	reel	PPA-CF	1	14.5	Reel shaft: 608 journals, head = clutch disk, D flat for the reel gear. Print head down

part	subsystem	material	qty	mass g	notes
reel_sprag_hub	reel	PPA-CF	1	44.4	Sprag hub: holds the CSK20PP backstop and bushes the capstan tube. Print front face down
reel_swing_arm	reel	PPA-CF	1	5.5	Swing arm: rides on the bus gear hub, carries the idler into mesh on forward wiper turns. Print flat

Printed mass: PLA 3835 g (1285.8 m of filament); PPA-CF 1435 g (477.2 m of filament).

Bought parts

item	type	count	used by
608ZZ 8x22x7	bearings	4	launch_aim_brg_1, launch_aim_brg_2, reel_brg_1, reel_brg_2
693ZZ 3x8x4	bearings	6	launch_muzzle_roller_brg, launch_pulley_top_brg, launch_pulley_bot_brg, launch_pulley_piv_brg, reel_p1_brg, reel_p2_brg
Adafruit AS5600 STEMMA QT PID 6357	bearings	1	sensors_as5600
CSK20PP 20x47x14 sprag #1	bearings	1	launch_sprag
CSK20PP one-way bearing 20x47x14	bearings	1	reel_sprag
M3 nut	fasteners	6	launch_sw_seated_nut, launch_sw_home_nut, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_sw_rail_low_nut, sensors_sw_gap_nut
M3 nut (jam)	fasteners	2	sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3 nyloc nut	fasteners	1	launch_muzzle_roller_axle
M3x10 self tap	fasteners	2	launch_guard_l_b2, launch_guard_r_b2
M3x10 self tap (lever clamp)	fasteners	6	reel_sw_lever, sensors_sw_rail_low_lever, sensors_sw_gap_lever
M3x10 SHCS	fasteners	2	sensors_sw_rail_low_screw, sensors_sw_gap_screw
M3x10 SHCS (foot into the housing)	fasteners	1	sensors_sw_rail_low_bracket
M3x10 SHCS + 2 M3 nuts	fasteners	1	reel_sw_hw
M3x10 SHCS + M3 nut	fasteners	1	reel_firm_screw
M3x12 self tap	fasteners	17	launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c
M3x12 SHCS	fasteners	2	sensors_as5600_strip_top, sensors_as5600_strip_bot
M3x12 SHCS (arm into the right cheek)	fasteners	2	sensors_sw_gap
M3x16 SHCS	fasteners	5	launch_muzzle_roller_axle, reel_bar_screws
M3x16 SHCS (deck screw)	fasteners	1	sensors_sw_rail_low_bracket
M3x16 SHCS (into the housing spine)	fasteners	2	sensors_as5600_carrier
M3x20 SHCS	fasteners	2	reel_crank_screw, reel_p1_axle
M3x20 SHCS + nyloc nut	fasteners	4	launch_pulley_top_axle, launch_pulley_bot_axle, launch_pulley_piv_axle, reel_p2_axle
M3x25 SHCS + M3 nut	fasteners	1	reel_rocker_pin
M3x6 SHCS	fasteners	2	launch_sw_seated_screw, launch_sw_home_screw
M3x8 self tap (base to bracket)	fasteners	8	reel_sw_base, sensors_sw_rail_low
M3x8 SHCS	fasteners	8	reel_hub_screws, reel_trim_screws, reel_sw_bracket
M8 hex nut ISO 4032	fasteners	1	launch_aim_nut
M8x25 hex bolt ISO 4017	fasteners	1	launch_aim_bolt
STEMMA QT JST SH 4 pin cable	fasteners	1	sensors_as5600
(part of the wiper motor)	other	1	frame_wiper_shaft

item	type	count	used by
12 V car wiper motor (self-locking worm), 2-wire rebuild	other	1	frame_wiper_motor
8 mm steel rod, cut to length (see note), ends deburred	other	2	launch_anchor_pin_l, launch_anchor_pin_r
BTS7960 43 A module (IBT-2)	other	1	frame_pcb_bts7960
Century C-243 (5/8 x 6-1/2 in x 0.054 in wire) extension spring	other	2	launch_spring_l_hook_a, launch_spring_r_hook_a
Dyneema braid 1 mm (LINE)	other	9	launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8
Espressif ESP32-S3-DevKitC-1-N8	other	1	frame_pcb_esp32
HX711 load cell amplifier (optional)	other	1	frame_pcb_hx711
inline ATO fuse holder + 20 A fuse	other	1	frame_fuse_20a
KEDGE v30 dart (reference, built separately)	other	1	launch_dart
launch_line_b (bought, no bought_ref)	other	1	launch_line_b
launch_line_c (bought, no bought_ref)	other	1	launch_line_c
launch_line_p (bought, no bought_ref)	other	1	launch_line_p
launch_spring_l_0 (bought, no bought_ref)	other	1	launch_spring_l_0
launch_spring_l_1 (bought, no bought_ref)	other	1	launch_spring_l_1
launch_spring_l_2 (bought, no bought_ref)	other	1	launch_spring_l_2
launch_spring_l_3 (bought, no bought_ref)	other	1	launch_spring_l_3
launch_spring_l_4 (bought, no bought_ref)	other	1	launch_spring_l_4
launch_spring_l_hook_c (bought, no bought_ref)	other	1	launch_spring_l_hook_c
launch_spring_r_0 (bought, no bought_ref)	other	1	launch_spring_r_0
launch_spring_r_1 (bought, no bought_ref)	other	1	launch_spring_r_1
launch_spring_r_2 (bought, no bought_ref)	other	1	launch_spring_r_2
launch_spring_r_3 (bought, no bought_ref)	other	1	launch_spring_r_3
launch_spring_r_4 (bought, no bought_ref)	other	1	launch_spring_r_4
launch_spring_r_hook_c (bought, no bought_ref)	other	1	launch_spring_r_hook_c
MP1584EN buck module, set to 9.0 V	other	1	frame_pcb_mp1584
N20 right-angle self-locking worm gearmotor 12 V 27 rpm (have)	other	1	drum_n20
perfboard 55 x 31 (dividers, pull-downs, 3V3 and GND buses), 2 x M3 x 10 on wall bosses	other	1	frame_perfboard
Pololu DRV8833 2130 (headers off)	other	1	frame_pcb_drv8833
Razer Kiyo X webcam	other	1	frame_kiyo_x

item	type	count	used by
RC832 5.8 GHz receiver (80 x 65 x 15)	other	1	frame_rc832
sensors_as5600_col_screws (bought, no bought_ref)	other	1	sensors_as5600_col_screws
sensors_as5600_nut_bot (bought, no bought_ref)	other	1	sensors_as5600_nut_bot
sensors_as5600_nut_top (bought, no bought_ref)	other	1	sensors_as5600_nut_top
sensors_as5600_screw_bot (bought, no bought_ref)	other	1	sensors_as5600_screw_bot
sensors_as5600_screw_top (bought, no bought_ref)	other	1	sensors_as5600_screw_top
sensors_sw_gap_jam (bought, no bought_ref)	other	1	sensors_sw_gap_jam
sensors_sw_rail_low_jam (bought, no bought_ref)	other	1	sensors_sw_rail_low_jam

4. Print list

151 printed parts on 17 plates. The 3MF plates and STL files are in the print pack.

part	material	qty	print size mm	supports
frame_deck_3	PLA	1	221 x 211 x 8	no
frame_deck_1	PLA	1	221 x 211 x 8	no
frame_deck_4	PLA	1	221 x 169 x 8	no
frame_deck_2	PLA	1	221 x 169 x 8	no
frame_camera_mount	PLA	1	58 x 72 x 236	yes, tree supports from the plate (672 mm2 over 45 deg)
frame_cheek_r	PLA	1	215 x 220 x 30	yes, tree supports from the plate (5227 mm2 over 45 deg)
launch_aim_housing	PLA	1	176 x 59 x 183	yes, tree supports from the plate (761 mm2 over 45 deg)
launch_rail_mid	PLA	1	124 x 68 x 170	yes, tree supports from the plate (2613 mm2 over 45 deg)
reel_wall	PLA	1	170 x 82 x 138	yes, tree supports from the plate (604 mm2 over 45 deg)
drum_drum_a	PLA	1	97 x 97 x 75	yes, tree supports from the plate (732 mm2 over 45 deg)
frame_motor_leg	PLA	1	49 x 69 x 187	yes, tree supports from the plate (222 mm2 over 45 deg)
launch_rail_rear	PLA	1	67 x 68 x 181	yes, tree supports from the plate (1750 mm2 over 45 deg)
launch_rail_front	PLA	1	132 x 128 x 48	yes, tree supports from the plate (7890 mm2 over 45 deg)
frame_ebay	PLA	1	124 x 124 x 58	yes, tree supports from the plate (879 mm2 over 45 deg)
drum_drum_b	PLA	1	60 x 60 x 58	yes, tree supports from the plate (1818 mm2 over 45 deg)
launch_stop_spring_l	PLA	1	155 x 52 x 14	no
launch_stop_spring_r	PLA	1	155 x 52 x 14	no
frame_ebay_lid	PLA	1	124 x 124 x 7	no
launch_guard_l_a	PLA	1	250 x 49 x 40	no

part	material	qty	print size mm	supports
launch_guard_r_a	PLA	1	250 x 49 x 40	no
launch_carriage	PLA	1	205 x 240 x 43	yes, tree supports from the plate (6453 mm2 over 45 deg)
drum_bracket_a	PLA	1	105 x 117 x 7	no
frame_can_stay	PLA	1	55 x 55 x 71	yes, tree supports from the plate (906 mm2 over 45 deg)
frame_guard_front	PLA	1	155 x 104 x 16	no
reel_spool	PLA	1	80 x 80 x 24	yes, tree supports from the plate (329 mm2 over 45 deg)
launch_stop_frame_l	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
launch_stop_frame_r	PLA	1	175 x 53 x 55	yes, tree supports from the plate (541 mm2 over 45 deg)
drum_bracket_b	PLA	1	79 x 117 x 6	no
frame_guard_hood	PLA	1	122 x 60 x 14	no
drum_fork	PLA	1	87 x 94 x 83	yes, tree supports from the plate (1302 mm2 over 45 deg)
launch_carriage_yoke	PLA	1	43 x 127 x 28	yes, tree supports from the plate (1406 mm2 over 45 deg)
reel_sw_bracket	PLA	1	40 x 63 x 58	yes, tree supports from the plate (334 mm2 over 45 deg)
frame_rc832_holder	PLA	1	97 x 22 x 40	no
drum_switch_block	PLA	1	26 x 51 x 24	yes, tree supports from the plate (216 mm2 over 45 deg)
launch_guard_r_b2	PLA	1	145 x 32 x 40	no
sensors_sw_rail_low_bracket	PLA	1	31 x 48 x 74	no
launch_guard_l_b2	PLA	1	145 x 32 x 40	no
drum_bracket_idler2	PLA	1	50 x 141 x 6	no
sensors_as5600_carrier	PLA	1	34 x 68 x 11	no
drum_bracket_idler	PLA	1	52 x 71 x 7	no
reel_shroud	PLA	1	81 x 80 x 41	no
reel_cup	PLA	1	48 x 52 x 18	yes, tree supports from the plate (1187 mm2 over 45 deg)
drum_n20_cradle	PLA	1	32 x 54 x 19	yes, tree supports from the plate (108 mm2 over 45 deg)
reel_spool_flange	PLA	1	80 x 80 x 2	no
sensors_sw_gap	PLA	1	63 x 43 x 12	no
drum_index_block	PLA	1	14 x 68 x 8	no
sensors_sw_rail_low	PLA	1	48 x 35 x 12	no
reel_sw_base	PLA	1	46 x 35 x 12	no
drum_dog_post	PLA	1	37 x 71 x 42	yes, tree supports from the plate (476 mm2 over 45 deg)
launch_aim_cap	PLA	1	46 x 14 x 19	no
launch_guard_l_c	PLA	1	32 x 54 x 26	no
launch_guard_r_c	PLA	1	32 x 54 x 26	no
launch_guard_l_b1	PLA	1	12 x 49 x 40	no
launch_guard_r_b1	PLA	1	12 x 49 x 40	no

part	material	qty	print size mm	supports
frame_cable_anchor_1	PLA	1	16 x 24 x 10	yes, tree supports from the plate (141 mm2 over 45 deg)
frame_cable_anchor_2	PLA	1	24 x 16 x 10	yes, tree supports from the plate (141 mm2 over 45 deg)
drum_hand_lever	PLA	1	46 x 24 x 6	no
drum_n20_strap	PLA	1	18 x 54 x 3	no
drum_magnet_cap	PLA	1	20 x 20 x 6	no
reel_t	PLA	1	26 x 26 x 5	no
frame_ebay_esp_clamp	PLA	1	43 x 8 x 4	no
sensors_sw_gap_lever	PLA	1	40 x 9 x 8	no
sensors_sw_rail_low_lever	PLA	1	44 x 5 x 8	no
reel_sw_lever	PLA	1	42 x 5 x 8	no
reel_pulley1	PLA	1	18 x 18 x 6	yes, tree supports from the plate (111 mm2 over 45 deg)
reel_pulley2	PLA	1	18 x 18 x 6	yes, tree supports from the plate (111 mm2 over 45 deg)
reel_bar_cap_1	PLA	1	10 x 20 x 5	no
reel_bar_cap_2	PLA	1	10 x 20 x 5	no
reel_c_clip	PLA	1	27 x 25 x 3	no
drum_sw_lever_0	PLA	1	14 x 48 x 6	no
drum_sw_lever_1	PLA	1	14 x 48 x 6	no
drum_sw_lever_2	PLA	1	14 x 48 x 6	no
sensors_as5600_strip_top	PLA	1	25 x 12 x 3	no
sensors_as5600_strip_bot	PLA	1	25 x 12 x 3	no
drum_index_lever	PLA	1	13 x 48 x 6	no
launch_sw_seated	PLA	1	42 x 8 x 6	no
launch_sw_home	PLA	1	42 x 8 x 6	no
launch_muzzle_roller	PLA	1	14 x 14 x 5	no
launch_pulley_top	PLA	1	14 x 14 x 5	no
launch_pulley_bot	PLA	1	14 x 14 x 5	no
launch_pulley_piv	PLA	1	14 x 14 x 5	no
launch_gear_washer	PLA	1	18 x 18 x 2	no
launch_aim_drag_pad	PLA	1	9 x 24 x 4	no
launch_gear_clip	PLA	1	17 x 18 x 1	no
reel_cup_clip	PLA	1	14 x 14 x 2	no
launch_rail_splice_l	PLA	1	22 x 5 x 3	no
launch_rail_splice_r	PLA	1	22 x 5 x 3	no
launch_drive_washer	PLA	1	32 x 32 x 1	no
reel_p2_clip	PLA	1	16 x 15 x 2	no
drum_shear_pin	PLA	6	12 x 5 x 4	no
reel_k_clip	PLA	1	14 x 13 x 1	no
launch_aim_drag_shim	PLA	1	9 x 7 x 4	no
launch_pulley_top_washer_l	PLA	1	9 x 9 x 2	no
launch_pulley_top_washer_r	PLA	1	9 x 9 x 2	no

part	material	qty	print size mm	supports
launch_pulley_bot_washer_l	PLA	1	9 x 9 x 2	no
launch_pulley_bot_washer_r	PLA	1	9 x 9 x 2	no
launch_aim_idler_clip	PLA	1	11 x 10 x 1	no
launch_aim_crankpin_clip	PLA	1	11 x 10 x 1	no
launch_aim_lugpin_clip	PLA	1	11 x 10 x 1	no
reel_s_clip	PLA	1	11 x 10 x 1	no
launch_seat_clip	PLA	1	6 x 6 x 2	no
launch_muzzle_washer_l	PLA	1	6 x 6 x 1	no
launch_muzzle_washer_r	PLA	1	6 x 6 x 1	no
frame_motor_adapter	PPA-CF	1	194 x 194 x 16	yes, tree supports from the plate (6734 mm2 over 45 deg)
launch_big_gear	PPA-CF	1	105 x 105 x 27	yes, tree supports from the plate (639 mm2 over 45 deg)
drum_gear_reel	PPA-CF	1	104 x 104 x 19	no
drum_coupler	PPA-CF	1	78 x 78 x 62	yes, tree supports from the plate (1998 mm2 over 45 deg)
reel_exit_post	PPA-CF	1	40 x 44 x 126	yes, tree supports from the plate (207 mm2 over 45 deg)
reel_p2_post	PPA-CF	1	102 x 29 x 125	yes, tree supports from the plate (1198 mm2 over 45 deg)
drum_shaft	PPA-CF	1	170 x 25 x 22	yes, tree supports from the plate (922 mm2 over 45 deg)
reel_drum	PPA-CF	1	46 x 46 x 43	yes, tree supports from the plate (251 mm2 over 45 deg)
launch_sleeve	PPA-CF	1	64 x 64 x 85	yes, tree supports from the plate (2907 mm2 over 45 deg)
drum_collar	PPA-CF	1	60 x 60 x 26	yes, tree supports from the plate (2694 mm2 over 45 deg)
reel_sprag_hub	PPA-CF	1	67 x 62 x 30	yes, tree supports from the plate (1504 mm2 over 45 deg)
launch_aim_crank	PPA-CF	1	62 x 60 x 38	yes, tree supports from the plate (985 mm2 over 45 deg)
launch_drive_gear	PPA-CF	1	70 x 70 x 29	yes, tree supports from the plate (2457 mm2 over 45 deg)
drum_idler_gear	PPA-CF	1	66 x 66 x 9	no
reel_c	PPA-CF	1	28 x 28 x 92	no
launch_aim_output	PPA-CF	1	48 x 48 x 60	no
reel_bus	PPA-CF	1	72 x 72 x 10	no
launch_aim_worm	PPA-CF	1	28 x 28 x 54	yes, tree supports from the plate (2779 mm2 over 45 deg)
drum_dogring_launch	PPA-CF	1	62 x 62 x 10	yes, tree supports from the plate (123 mm2 over 45 deg)
launch_aim_sun	PPA-CF	1	48 x 48 x 12	no
reel_gc	PPA-CF	1	48 x 48 x 13	no
launch_aim_lug	PPA-CF	1	30 x 58 x 28	yes, tree supports from the plate (995 mm2 over 45 deg)
drum_coupler_hub	PPA-CF	1	26 x 26 x 32	no
reel_s	PPA-CF	1	62 x 62 x 56	no

part	material	qty	print size mm	supports
launch_aim_conrod	PPA-CF	1	173 x 17 x 6	no
drum_axle	PPA-CF	1	141 x 10 x 9	no
drum_rail	PPA-CF	1	141 x 10 x 9	no
reel_k	PPA-CF	1	10 x 10 x 126	no
reel_gi	PPA-CF	1	48 x 48 x 6	no
reel_gk	PPA-CF	1	39 x 39 x 11	yes, tree supports from the plate (668 mm2 over 45 deg)
launch_aim_bevel30	PPA-CF	1	53 x 52 x 10	yes, tree supports from the plate (590 mm2 over 45 deg)
reel_bar	PPA-CF	1	154 x 19 x 16	no
drum_idler	PPA-CF	1	22 x 22 x 46	yes, tree supports from the plate (331 mm2 over 45 deg)
launch_aim_idler	PPA-CF	1	30 x 30 x 12	no
launch_aim_arm	PPA-CF	1	64 x 41 x 19	no
reel_rocker	PPA-CF	1	29 x 92 x 6	no
reel_swing_arm	PPA-CF	1	84 x 38 x 9	no
launch_gear_pin	PPA-CF	1	53 x 16 x 12	no
reel_gr	PPA-CF	1	33 x 33 x 6	no
reel_p2_fork	PPA-CF	1	19 x 14 x 36	yes, tree supports from the plate (133 mm2 over 45 deg)
reel_crank	PPA-CF	1	21 x 28 x 12	no
launch_yoke_peg_l	PPA-CF	1	20 x 8 x 7	no
launch_yoke_peg_r	PPA-CF	1	20 x 8 x 7	no
launch_seat_plunger	PPA-CF	1	30 x 15 x 20	no
drum_reel_washer	PPA-CF	1	35 x 35 x 1	no
drum_n20_pinion	PPA-CF	1	10 x 9 x 8	no
launch_seat_pushrod	PPA-CF	1	16 x 5 x 4	no
reel_crank_washer	PPA-CF	1	12 x 12 x 1	no

plate_1_PLA.3mf: launch_carriage, launch_guard_l_b2, sensors_as5600_carrier.

plate_2_PLA.3mf: drum_switch_block, frame_cable_anchor_2, frame_cheek_r, reel_t, sensors_as5600_strip_bot, sensors_as5600_strip_top.

plate_3_PLA.3mf: drum_magnet_cap, frame_deck_3, launch_gear_clip, launch_gear_washer, launch_pulley_bot, launch_pulley_piv, launch_pulley_top, reel_bar_cap_1, reel_bar_cap_2, reel_p2_clip, reel_pulley1, reel_pulley2.

plate_4_PLA.3mf: frame_deck_1, reel_k_clip.

plate_5_PLA.3mf: drum_bracket_idler, frame_deck_4, launch_aim_housing.

plate_6_PLA.3mf: drum_shear_pin__c4, drum_shear_pin__c5, drum_shear_pin__c6, drum_sw_lever_0, drum_sw_lever_1, drum_sw_lever_2, frame_can_stay, frame_deck_2, launch_aim_drag_pad, launch_guard_l_b1, launch_muzzle_washer_l, launch_muzzle_washer_r, launch_pulley_top_washer_l, launch_pulley_top_washer_r, launch_seat_clip, launch_stop_frame_l, reel_s_clip, reel_sw_lever.

plate_7_PLA.3mf: drum_index_lever, frame_guard_front, frame_rc832_holder, launch_pulley_bot_washer_r, launch_rail_front, launch_rail_rear, launch_rail_splice_l, launch_rail_splice_r, reel_cup_clip, reel_wall.

plate_8_PLA.3mf: drum_bracket_a, drum_bracket_b, drum_bracket_idler2, drum_drum_a, drum_index_block, drum_shear_pin__c1, frame_ebay, launch_muzzle_roller.

plate_9_PLA.3mf: frame_camera_mount, frame_ebay_lid, launch_aim_drag_shim, launch_carriage_yoke, launch_pulley_bot_washer_l, launch_rail_mid, launch_stop_frame_r, launch_stop_spring_l, sensors_sw_gap, sensors_sw_gap_lever.

plate_10_PLA.3mf: drum_drum_b, drum_fork, drum_shear_pin__c2, drum_shear_pin__c3, frame_guard_hood, frame_motor_leg, launch_aim_crankpin_clip, launch_aim_lugpin_clip, launch_guard_r_b2, launch_stop_spring_r, reel_shroud, reel_spool, sensors_sw_rail_low_lever.

plate_11_PLA.3mf: drum_dog_post, drum_hand_lever, drum_n20_cradle, drum_n20_strap, frame_cable_anchor_1, frame_ebay_esp_clamp, launch_aim_cap, launch_aim_idler_clip, launch_drive_washer, launch_guard_l_c, launch_guard_r_b1, launch_guard_r_c, launch_sw_home, launch_sw_seated, reel_c_clip, reel_cup, reel_spool_flange, reel_sw_base, reel_sw_bracket, sensors_sw_rail_low, sensors_sw_rail_low_bracket.

plate_12_PLA.3mf: launch_guard_l_a.

plate_13_PLA.3mf: launch_guard_r_a.

plate_14_PPA-CF.3mf: drum_axle, drum_reel_washer, frame_motor_adapter, reel_bar, reel_drum, reel_exit_post, reel_gr, reel_p2_fork, reel_swing_arm.

plate_15_PPA-CF.3mf: drum_coupler, drum_coupler_hub, drum_gear_reel, drum_n20_pinion, launch_aim_bevel30, launch_aim_conrod, launch_aim_output, launch_aim_sun, launch_big_gear, launch_gear_pin, launch_seat_pushrod, launch_yoke_peg_l, launch_yoke_peg_r, reel_bus, reel_crank, reel_gi, reel_k.

plate_16_PPA-CF.3mf: drum_collar, drum_dogring_launch, drum_idler_gear, drum_shaft, launch_aim_arm, launch_aim_crank, launch_aim_lug, launch_drive_gear, launch_seat_plunger, launch_sleeve, reel_gk, reel_p2_post, reel_rocker, reel_s, reel_sprag_hub.

plate_17_PPA-CF.3mf: drum_idler, drum_rail, launch_aim_idler, launch_aim_worm, reel_c, reel_crank_washer, reel_gc.

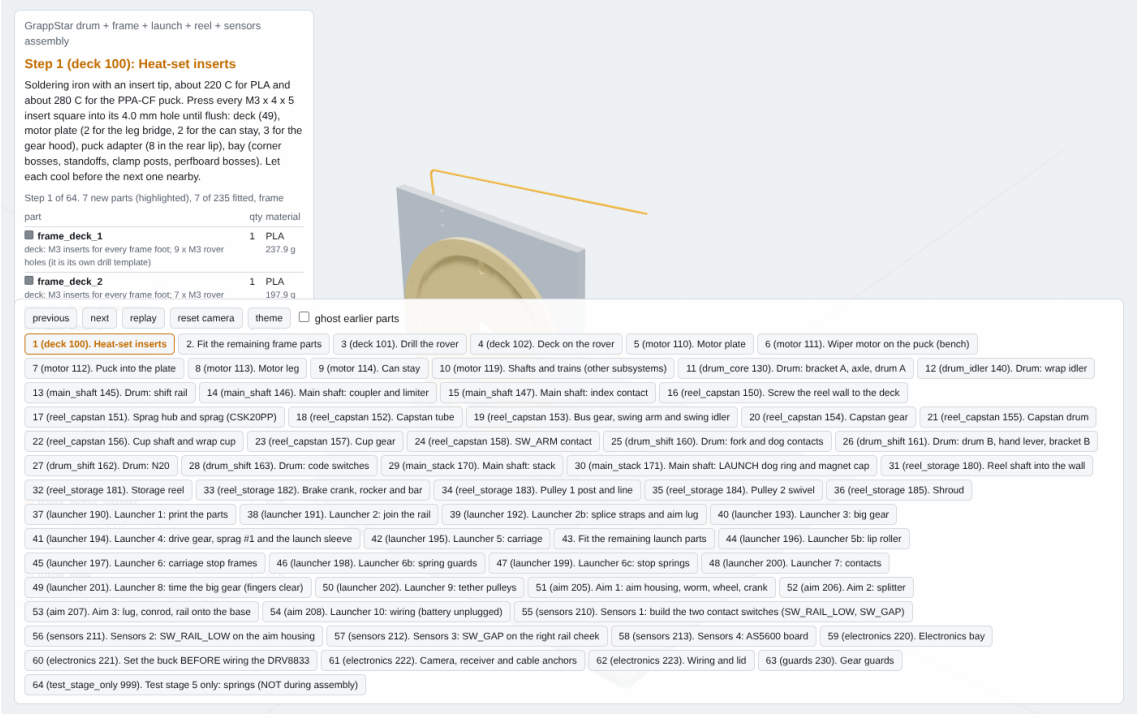
Print settings are in PRINT_SHEET.pdf. Use a hardened nozzle for PPA-CF and dry it first.

5. Assembly steps

Step 1 (deck 100). Heat-set inserts

Battery unplugged. Soldering iron with an insert tip, about 220 C for PLA and about 280 C for the PPA-CF puck. Press every M3 x 4 x 5 insert square into its 4.0 mm hole until flush: deck (49), motor plate (2 for the leg bridge, 2 for the can stay, 3 for the gear hood), puck adapter (8 in the rear lip), bay (corner bosses, standoffs, clamp posts, perfboard bosses). Let each cool before the next one nearby.

Parts: frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4, frame_cheek_r, frame_motor_adapter, frame_ebay.



Step 2. Fit the remaining frame parts

Parts without an explicit assembly step: frame_perfboard.

Parts: frame_perfboard.

Step 3 (deck 101). Drill the rover

Lay the deck on the rover top plate where it will live (launch direction forward). Mark through the 32 perimeter holes with a 3 mm drill or a scriber (or tape down the 1:1 PDF template from cad/parts/_frame_tests/drill_template.py), take the deck off, drill 3.4 mm. Deburr.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

Step 4 (deck 102). Deck on the rover

Slide the tiles together on the dovetails (from above). M3 x 25 SHCS down through every perimeter hole, washer and nyloc under the rover plate. Snug, not crushed. Do this before the other subsystems put their feet down: some rover bolts sit under their feet.

Also handle (fitted earlier): frame_deck_1, frame_deck_2, frame_deck_3, frame_deck_4.

Step 5 (motor 110). Motor plate

Hold the motor plate upright (front face, the side without the bridge, towards +y) with its bridge against the top of the leg. 2 x M3 x 12 along y through the leg's long counterbores into the bridge inserts.

Also handle (fitted earlier): frame_cheek_r.

Step 6 (motor 111). Wiper motor on the puck (bench)

Battery unplugged. Put the M6 bolts through the puck slots from the front (heads sit in the slot counterbores), into the motor bosses (threaded bosses, or a nut behind each boss ear). Slide the motor so its shaft is centred in the bore (a 10 mm drill shank in the coupler bore helps), then tighten.

Parts: frame_wiper_motor, frame_wiper_shaft.
Also handle (fitted earlier): frame_motor_adapter.

Step 7 (motor 112). Puck into the plate

Feed the puck into the plate window from behind (the motor side) until its lip sits on the plate's rear face. From the front, M3 x 12 through the plate into the lip inserts (5 of the 8 rim holes have both plate and lip; use those).

Also handle (fitted earlier): frame_motor_adapter.

Step 8 (motor 113). Motor leg

Stand the leg on the deck in the corner between the reel and the bay, foot along y. 4 x M3 x 10 down through the foot into the deck inserts.

Parts: frame_motor_leg.

Step 9 (motor 114). Can stay

Post against the plate's rear face, 2 x M3 x 12 into the plate inserts. Cable tie or hose clamp round the motor can through the two cradle slots, firm but not crushing.

Parts: frame_can_stay.

Step 10 (motor 119). Shafts and trains (other subsystems)

Now cad-drum (coupler on the wiper shaft, SW_INDEX block on the strut of idler post 2: drum step "Main shaft: index contact"), cad-reel and cad-launch install their trains and aim housing. Follow their steps, then come back. The tether crosses the puck twice, in channels cut out to the puck's edge: from the guide eye to the cup ring it runs in the channel that opens into the capstan opening (lay it in from there); from pulley 1 forward to pulley 2 it runs in the notch near the rim, which the plate closes (push the line end through it from behind).

Step 11 (drum_core 130). Drum: bracket A, axle, drum A

Bolt bracket A to the deck (2 x M3 x 12). Press the drum axle into it (flat facing forward). Slide drum A onto the axle, gear first, and turn it so the gap in its lock ring faces back toward the idler position (about drum 180 on the pointer). The shift rail goes in after the idler.

Parts: drum_bracket_a, drum_axle, drum_drum_a.

Step 12 (drum_idler 140). Drum: wrap idler

Push the idler output gear onto the idler shaft flat from the long end. Slide idler post 2 onto the short journal and idler post 1 onto the long journal (the thick middle of the shaft sits between the posts and locates it). Fit this before cad-reel's bus gear, SW_ARM bracket and cup gear. Lower the posts with the idler onto the deck about 4 mm to the back (-x) of their place, then slide them forward so the small pinion meshes the drum A segment teeth, and bolt them down (2 x M3 x 12 each; the front screw of post 2 needs the short arm of the hex key). Turn drum A back to LAUNCH (mark L at the top): the flat-sided lock cam must sit in the drum lock ring and the idler can no longer turn by hand.

Parts: drum_idler_gear, drum_idler, drum_bracket_idler, drum_bracket_idler2.

Step 13 (main_shaft 145). Drum: shift rail

Slide the shift rail in from the outboard end, over the idler posts, and press it into bracket A (flat facing forward).

Parts: drum_rail.

Step 14 (main_shaft 146). Main shaft: coupler and limiter

Push the coupler hub onto the wiper output shaft (flats in line) until it stops on the end of the flats; its lower end goes into the puck bore, just clear of the motor's output boss. Drop the M8 nut into the hub pocket onto the thread (it sits about 8 mm down the pocket) and tighten it with a 13 mm socket (1/4 inch drive). Slide the outer coupler over the hub, line up the holes, push a shear pin in (head outside, its flat facing the outboard end) and put a piece of tape round the coupler over the head. Print the pins lying on their flat (D44: shear across the layers). Keep the 5 spare pins with the base. A broken pin stub comes out with an M3 screw turned into it. Then slide the REEL gear onto the coupler journal (plain face first) and screw the retaining washer to the journal end (2 x M3 x 6): the gear must spin free (it cannot go on once the dog post is fitted).

Parts: drum_coupler_hub, drum_coupler, drum_shear_pin, drum_gear_reel, drum_reel_washer.

Step 15 (main_shaft 147). Main shaft: index contact

Battery unplugged. Press an M3 nut into the index block pocket, screw the index lever foot to the block (M3 x 12), thread an M3 x 6 into the lever boss, head toward the nut. Screw the block to the inner face of the strut of idler post 2 (2 x M3 x 12 from the far side of the post into the block), with the nub on the coupler cam. Set the gap so the screw head touches the nut near the top of a bump (about 1.5 mm on a low part). Battery unplugged: nut to GND, screw to SW_INDEX.

Parts: drum_index_block, drum_index_lever.

Step 16 (reel_capstan 150). Screw the reel wall to the deck

Stand the reel wall on the deck at its marked place. Put seven M3 screws down through the counterbored foot holes into the deck inserts and tighten. Do this first: the storage reel and the gears later cover some of the holes.

Parts: reel_wall.

Step 17 (reel_capstan 151). Sprag hub and sprag (CSK20PP)

Push the sprag hub's round spigot into the big hole in the wall from the front, flange flat on the wall, and screw it on from the front with four M3 x 8 screws (they tap into the wall). Find the key slot on the outside of the CSK20PP and line it up with the printed rib at the top of the hub's seat. Check the arrow on the bearing: seen from the front, the inner ring must turn freely counter-clockwise and lock clockwise. Press it in flat until it sits on the lip.

Parts: reel_sprag_hub, reel_hub_screws, reel_sprag.

Step 18 (reel_capstan 152). Capstan tube

Slide the capstan tube in from the FRONT, D-flat end first, its key rib lined up with the key slot inside the sprag. Push until the collar sits on the sprag. It turns one way only (counter-clockwise seen from the front).

Parts: reel_c.

Step 19 (reel_capstan 153). Bus gear, swing arm and swing idler

From behind the wall slide the 46-tooth bus gear onto the tube, gear face first, hub toward you. Slip the swing arm's ring over the bus gear's hub, idler pin pointing at the wall, and put the 30-tooth swing idler on the pin. The idler is held between the arm and the wall. The arm swings 12 degrees between its two stop pins.

Parts: reel_bus, reel_swing_arm, reel_gi.

Step 20 (reel_capstan 154). Capstan gear

Slide the 22-tooth capstan gear onto the D-flat behind the bus gear, hub toward the bus gear, and snap the large C-clip into the groove behind it. The bus gear must still turn freely.

Parts: reel_gc, reel_c_clip.

Step 21 (reel_capstan 155). Capstan drum

Slide the green drum onto the front of the tube, flange first, key slot on the rib, until it sits on the collar.

Parts: reel_drum.

Step 22 (reel_capstan 156). Cup shaft and wrap cup

Push the cup shaft through the tube from the FRONT, plain end first (the neck end stays in front), until its back end comes out behind the capstan gear. Slide the wrap cup onto the front end (flat in the bore onto the flat of the shaft, ring at the top), over the drum. Push Harry's small C-clip sideways onto the neck in front of the cup.

Parts: reel_k, reel_cup, reel_cup_clip.

Step 23 (reel_capstan 157). Cup gear

From behind, slide the 24-tooth cup gear onto the D-flat at the back of the cup shaft, hub forward, and snap its C-clip on. The cup must turn freely both ways.

Parts: reel_gk, reel_k_clip.

Step 24 (reel_capstan 158). SW_ARM contact

Battery unplugged. Build the printed contact (see the contact switch page): nut pressed in the base trap, lever screwed onto the pad with 2 x M3 x 10 self tapping screws, M3 x 10 contact screw with its jam nut in the lever. Screw the switch base to the bracket with 4 x M3 x 8 self tapping screws (into the pilot holes in the plate), and the bracket to the back of the wall below the gears with two M3 x 8. SW_ARM is closed while the cup is away from home and opens at home: the lever rides a raised track on the back of the cup gear and drops into a notch in it at home. Turn the cup to home (ring at the top): the lever tip sits free in the notch. Turn the contact screw until its head just touches the nut, then back it off 1.9 turns (gap about 0.9 mm, 0.5 mm per turn) and lock the jam nut. Check with the meter: turn the cup about 15 degrees either way, the lever rides up onto the track and the contact closes; back at home it must open, for about 7 degrees either side.

Parts: reel_sw_bracket, reel_sw_base, reel_sw_lever, reel_sw_hw.

Step 25 (drum_shift 160). Drum: fork and dog contacts

Battery unplugged. Press an M3 nut into the hex hole of the yoke tab. Thread an M3 x 8 into each dog post boss first, heads toward each other (M3 x 8: a longer screw reaches the REEL gear; no room for a driver later). Turn the drum to about 180. Slide the fork sleeve onto the rail, yoke facing the main shaft, and swing it about 20 deg pin end away from the drum while the dog post slides on (its bar reaching back over the yoke tab; fit it before cad-reel's pulley 2). Grease the rail. Tighten the dog post clamp screw (M3 x 8, tapped) onto the rail flat. Battery unplugged: nut wire to GND, REEL side screw to SW_DOG_R, LAUNCH side screw to SW_DOG_L.

Parts: drum_dog_post, drum_fork.

Step 26 (drum_shift 161). Drum: drum B, hand lever, bracket B

Swing the fork up on the rail, slide drum B onto the axle (hex socket over the drum A spigot; it fits one way only, cut corner to filled corner), slide the fork to its middle position and swing it down so its pin drops into the groove (drum near 180). Slide the hand lever onto the hex boss. Press bracket B onto the axle and rail ends and bolt it down (2 x M3 x 12). The drum turns by hand with about 0.6 mm end play.

Parts: drum_drum_b, drum_hand_lever, drum_bracket_b.

Step 27 (drum_shift 162). Drum: N20

Battery unplugged. Press the pinion onto the N20 output shaft (flat to flat) leaving 1.5 mm of shaft below it (that end runs in the bracket A ear slot). Slide the N20 into the cradle from the open end, fit the end plate (2 x M3 x 8). Press an M3 nut into the pocket in the bracket A ear front face, wire along its groove. Screw the cradle to the ear from the cradle side: pivot screw M3 x 12 through the round hole, swing screw M3 x 12 through the arc slot at its full-mesh end (the slot only lets the cradle swing out), both tapped into the ear. Turn the SW_N20 screw (M3 x 8, tapped in the cradle plate) in until its tip just touches the nut. Battery unplugged: nut to GND, screw to SW_N20.

Parts: drum_n20_pinion, drum_n20, drum_n20_cradle, drum_n20_strap.

Step 28 (drum_shift 163). Drum: code switches

Battery unplugged. Press an M3 nut into each of the 3 front pockets of the switch block and bolt it to the deck (2 x M3 x 12). Screw each lever foot to the block (M3 x 12), nub on its band. Thread an M3 x 6 into each lever boss, head toward the nut, so the head touches the nut near the top of a bump (gap about 1.8 mm with the nub low). Battery unplugged: nuts to GND, screws to SW_DRUM_A (nearest the motor), SW_DRUM_B, SW_DRUM_C.

Parts: drum_switch_block, drum_sw_lever_0, drum_sw_lever_1, drum_sw_lever_2.

Step 29 (main_stack 170). Main shaft: stack

Slide the collar onto the main shaft from the spigot end. Lower the shaft with its hex end outboard of the REEL gear dogs (beyond s 70) so the collar groove drops between the fork pads, then push it about 42 mm into the coupler socket. The shaft is held endwise later by the launch sleeve and the magnet cap.

Parts: drum_shaft, drum_collar.

Step 30 (main_stack 171). Main shaft: LAUNCH dog ring and magnet cap

Battery unplugged. Fit the LAUNCH dog ring on the launch sleeve flange (sleeve lugs in the ring slots, 3 x M3 x 10 from the sleeve side into the ring) before the launch sleeve goes on the spigot (launcher steps). After the launcher: glue the AS5600 magnet into the cap, push the cap on the spigot end (flat to flat) and fit the M3 x 6 onto the flat. Hand use of the drum: battery unplugged, slacken the line, loosen the swing screw, swing the N20 out, turn the lever, swing back and tighten.

Parts: drum_dogring_launch, drum_magnet_cap.

Step 31 (reel_storage 180). Reel shaft into the wall

Feed the reel shaft's thin end through the rocker's hole and the wall bushing from the front. From behind, slide the 20-tooth reel gear onto the D-flat, hub toward the wall, and snap its C-clip on. Swing the arm toward the reel gear (to its stop): the idler meshes it. Spin the reel by hand: it should turn with a light, even drag. Set LIGHT (after the line is wound on, drum at LAUNCH): swing the arm to its parked stop (the idler leaves the reel gear), turn the reel shaft until the reel gear's two small holes line up with the wall holes, swing the arm back, reach through with a 2.5 mm hex key and turn both trim screws the same amount until a 50 g weight hung on the line end just pulls line off slowly (0.5 N). 1/8 turn changes it by about 0.4 N of squeeze.

Parts: reel_gr, reel_s_clip.

Step 32 (reel_storage 181). Storage reel

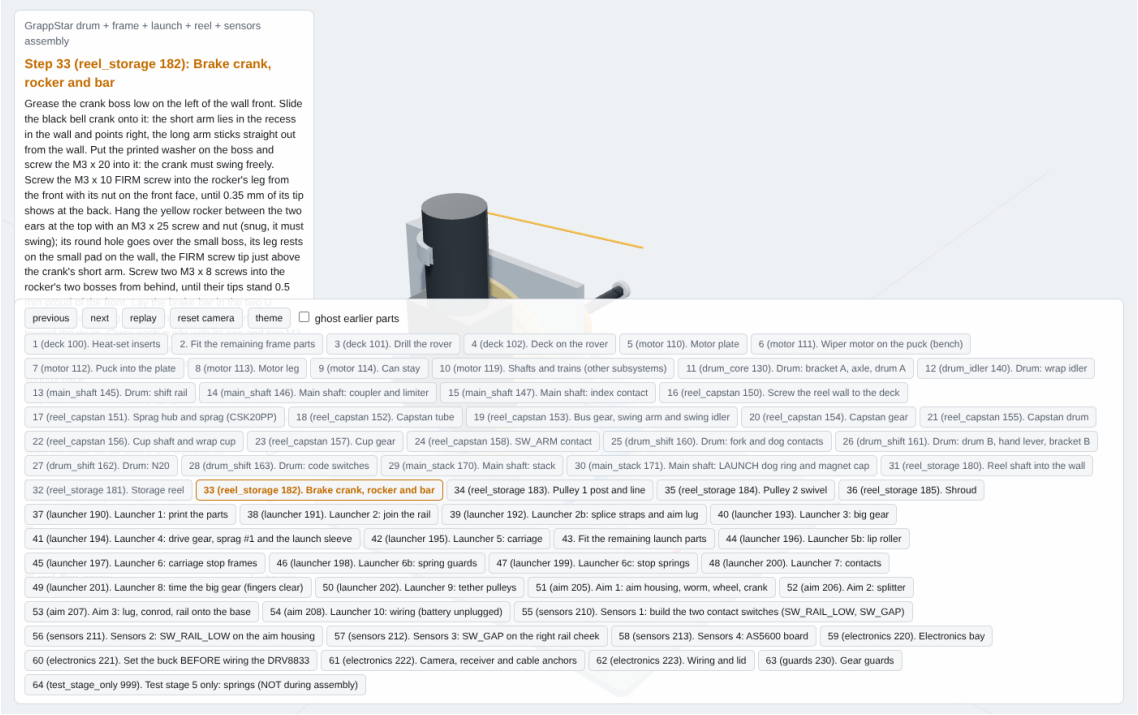
Press one 608 bearing into each end of the reel body. Push the Dyneema end through the small hole in the core near the back flange into the open pocket, tie a stopper knot (figure eight) and pull it tight. Lay the flat flange disc on the front, its slots on the three lugs, raised ring outward. Slide the reel onto the reel shaft from the shaft's thin end until the disc meets the shaft's round head. Slide the thrust ring onto the shaft behind the reel, small boss toward the bearing.

Parts: reel_spool, reel_spool_flange, reel_brg_1, reel_brg_2, reel_s, reel_t.

Step 33 (reel_storage 182). Brake crank, rocker and bar

Grease the crank boss low on the left of the wall front. Slide the black bell crank onto it: the short arm lies in the recess in the wall and points right, the long arm sticks straight out from the wall. Put the printed washer on the boss and screw the M3 x 20 into it: the crank must swing freely. Screw the M3 x 10 FIRM screw into the rocker's leg from the front with its nut on the front face, until 0.35 mm of its tip shows at the back. Hang the yellow rocker between the two ears at the top with an M3 x 25 screw and nut (snug, it must swing); its round hole goes over the small boss, its leg rests on the small pad on the wall, the FIRM screw tip just above the crank's short arm. Screw two M3 x 8 screws into the rocker's two bosses from behind, until their tips stand 0.5 mm proud of the front. Lay the brake bar in the two U guides, its raised lug against the crank's long arm, nose toward the drum. Close each guide with its cap and two M3 x 16 screws. Push the nose: the crank swings and the rocker leg moves away from the wall; let go and it all springs back.

Parts: reel_crank, reel_crank_washer, reel_crank_screw, reel_firm_screw, reel_rocker, reel_rocker_pin, reel_trim_screws, reel_bar, reel_bar_cap_1, reel_bar_cap_2, reel_bar_screws.



Step 34 (reel_storage 183). Pulley 1 post and line

Battery unplugged. Stand the pulley 1 post on the deck in front of the wall and screw it down with four M3 screws into the deck. Press a 693ZZ into pulley 1 and set it between the post's two cheeks. Wind the line onto the reel (counter-clockwise seen from the front, line over the top), thread it through the guide eye, over the top of the cup and through the cup ring from the back. Lay it over pulley 1 in its groove (it comes up from the ring and turns down toward the deck), then push the M3 x 20 axle down through the top cheek and the pulley and screw it into the post below (threadlocker, no nut). Lead the line down to pulley 2. Set FIRM last, with the drum in REEL (the bar pushed in by the cam): pull the line end with the luggage scale; it must slip at 5.0 N with the reel full. Too low: loosen the lock nut on the rocker leg and turn the FIRM screw in (toward the wall); too high: out. One turn is about 1 N at the line. Lock the nut.

Parts: reel_exit_post, reel_pulley1, reel_p1_brg, reel_p1_axle.

Step 35 (reel_storage 184). Pulley 2 swivel

Press a 693ZZ into pulley 2, set it between the swivel fork's two cheeks and push the M3 x 20 axle through (nyloc nut on the far side, snug: the pulley must spin). Grease the fork's tube and push it through the bore of the pulley 2 post's block from the pulley side (the tube is level and points toward pulley 1); snap the C-clip into the groove where the tube comes out of the block. The fork must turn freely. Screw the post to the deck beside the drum with four M3 screws into the deck inserts. The line from pulley 1 goes through the tube, round the pulley and out toward the rail pivot pulley; the fork turns itself to follow the line as the rail aims.

Parts: reel_p2_post, reel_p2_fork, reel_p2_clip, reel_pulley2, reel_p2_brg, reel_p2_axle.

Step 36 (reel_storage 185). Shroud

Put the shroud round the reel, its flange on the wall, opening at the top right, and screw it on with three M3 x 8 screws into the wall. The reel must not touch it anywhere.

Parts: reel_shroud.

Step 37 (launcher 190). Launcher 1: print the parts

All PLA unless named. PPA-CF: big gear, drive gear, launch sleeve, big gear pin, SW_SEATED plunger and pushrod, and the aim gears (sun, idler, output, bevel, worm, crank). Use each part's print orientation from the print sheet: the carriage body lies on its side across the bed diagonal; the stop springs print upside down at 100 percent infill (all perimeters); rail pieces lie on their largest flat face. 0.2 mm layers, 4 walls on every loaded part. Hardware (launcher and aim train): M3 screws 2 x 6, 3 x 8, 14 x 10, 23 x 12, 5 x 16, 15 x 20, 4 x 30, plus 4 x 10 for the aim housing to the deck (Aim 3); 10 M3 nuts, 4 M3 nylocs; M8 x 25 bolt and M8 nut; 2 x 608; 4 x 693ZZ (3 tether pulleys, 1 lip roller); 2 x 8 mm steel rod pins (cut to the length on the print sheet) for the spring anchors; sprag #1 (CSK20PP); the two springs.

Step 38 (launcher 191). Launcher 2: join the rail

Lay the rear, mid and front rail pieces in a line on a flat table. Push each joint together and fit 2 x M3 x 20 with nuts through each flange pair (8 in total). Tighten evenly; file any step inside the track where the pieces meet so the carriage shoes slide over it.

Parts: launch_rail_rear, launch_rail_mid, launch_rail_front.

Step 39 (launcher 192). Launcher 2b: splice straps and aim lug

Screw a splice strap on each wall top across the front joint (4 x M3 x 10 self tap each). Bolt the aim lug to the left rail wall now (2 x M3 x 12 self tap): the spring guards cover its screws later.

Parts: launch_rail_splice_l, launch_rail_splice_r, launch_aim_lug.

Step 40 (launcher 193). Launcher 3: big gear

Hold the big gear between the two cheeks of the mid piece, the toothed sector towards the drum side (right) and the full ring of teeth towards the left, with the spacer washer between the gear and the left cheek. Push the big gear pin through cheek, gear, washer, cheek from the right and snap the C clip into its groove. The gear must spin freely with almost no end play.

Parts: launch_big_gear, launch_gear_pin, launch_gear_clip, launch_gear_washer.

Step 41 (launcher 194). Launcher 4: drive gear, sprag #1 and the launch sleeve

First bolt cad-drum's LAUNCH dog ring to the sleeve flange: lugs in the ring slots, 3 x M3 x 8 from the flange side into the tapped ring (the heads are out of reach once the sleeve is in the cheeks). Slide sprag #1 into the drive gear so its outer keyway takes the rib inside the gear. Check the direction: look at the gear from its collar side (the side that goes to the right, towards the motor), hold the gear and turn the inner race with a finger. It must turn freely counter-clockwise (BACKWARD) and lock clockwise (FORWARD: clockwise seen from the motor side is the cock direction). If it is the other way round, push the sprag out, turn it over and slide it in other face first. Hold the drive gear between the cheeks (collar side right, then the spacer washer on its left), meshing the big gear's full ring of teeth. Turn the cheek bores so their slots point down, then push the launch sleeve in from the motor side, outboard end first, ribs lined up with the slots and the sprag keyway, until the long (15 mm) rib sits inside the sprag.

Parts: launch_drive_gear, launch_sprag, launch_sleeve, launch_drive_washer.

Step 42 (launcher 195). Launcher 5: carriage

Turn the big gear so its toothless gap faces the rack. Slide the carriage BODY into the track from the FRONT end (the tether fork blocks the rear), rack teeth down, until its rear face reaches the stop position. Yoke: push the SW_SEATED pushrod into the block bore from the rear, collar first, then push the plunger's pin up through the block from below (pad under the left arm) and press the head cap on the pin top. Lower the yoke onto the carriage rear so its key plates drop into the slots in the body webs, and fit 1 x M3 x 10 per side through the yoke into the webs. Press a printed spring peg up into each inner (y 38) peg hole from below, flat to the rear, to the end of the hole.

Parts: launch_carriage, launch_carriage_yoke, launch_seat_plunger, launch_seat_pushrod, launch_seat_clip, launch_yoke_peg_l, launch_yoke_peg_r.

Step 43. Fit the remaining launch parts

Parts without an explicit assembly step: launch_dart.

Parts: launch_dart.

Step 44 (launcher 196). Launcher 5b: lip roller

Press a 693ZZ into the lip roller, put a printed thrust washer each side in its bore (hub to the bearing), drop roller and washers into the fork at the carriage front and fit the M3 x 16 and a nyloc nut; it must spin freely. The tether runs over it in tow and retrieval, and the dart's tail slides in over it through the funnel onto the cradle.

Parts: launch_muzzle_roller, launch_muzzle_roller_brg, launch_muzzle_roller_axle, launch_muzzle_washer_l, launch_muzzle_washer_r.

Step 45 (launcher 197). Launcher 6: carriage stop frames

For each side, before any spring: slide the stop frame onto the rail from the outboard side (it does not drop straight down past the inboard preload ledge). 2 x M3 x 12 through the rear foot into the rail wall, and 2 x M3 x 16 down through the front foot into the rail front plate, driven through the access holes in the frame's front wall.

Parts: launch_stop_frame_l, launch_stop_frame_r.

Step 46 (launcher 198). Launcher 6b: spring guards

Screw the spring guard plates to the rail walls and anchor posts (M3 self tap into the pilots, see the print sheet), four per side: a (rear, with the end wall), b1 and b2 (either side of the stop frame's rear leg) and c (front, 2 x M3 x 12 into the front plate edge). Slide each one in from the side, under the stop frame's outboard bar. Wall brackets: M3 x 12 self tap, head and hex key through the hole in the plate; b2 foot: M3 x 10 down. Never cock with a guard off.

Parts: launch_guard_l_a, launch_guard_l_b1, launch_guard_l_b2, launch_guard_l_c, launch_guard_r_a, launch_guard_r_b1, launch_guard_r_b2, launch_guard_r_c.

Step 47 (launcher 199). Launcher 6c: stop springs

Put each stop spring on its ledges inside its frame, rear bar towards the yoke arm, and pull it onto the frame's front wall with the 2 x M3 x 20 draw screws (heads on the front wall), a turn each in turn, until it sits on the wall (6.6 mm of preload). The rear bar must rest on both ledges.

Parts: launch_stop_spring_l, launch_stop_spring_r.

Step 48 (launcher 200). Launcher 7: contacts

Battery unplugged. SW_HOME (right lip) and SW_SEATED (left lip): drop an M3 nut into the slot in the rail wall with a stripped wire end under it. Put a ring lug (or a stripped wire end wrapped once) on an M3 x 6 under its head and screw it up into the lever's boss, head down, until the tip is flush with the boss top (the pad beside it must stay the highest point), then screw the lever root to the lip with one M3 x 10 (self tap). Set the gap with the carriage pushed back at least 10 mm (lever free): turn the screw down until the head just touches the nut (meter beeps), then back it off ONE full turn (0.5 mm gap; a dab of nail varnish on the thread holds it). Carriage on its stop: the right foot presses the pad about 0.95 mm and SW_HOME must read closed; 2 mm back it must read open (if not, a quarter turn at a time). SW_SEATED closes only with a dart on the sled (no springs yet: hold the carriage on its stop by hand to check). SW_GAP, SW_RAIL_LOW and the AS5600 carrier are fitted in the Sensors steps (cad-sensors).

Parts: launch_sw_home, launch_sw_home_screw, launch_sw_home_nut, launch_sw_seated, launch_sw_seated_screw, launch_sw_seated_nut.

Step 49 (launcher 201). Launcher 8: time the big gear (fingers clear)

No springs fitted yet. Hold the carriage on its stop by hand and turn the drive gear FORWARD until the first tooth after the toothless gap just touches the rearmost rack tooth: this is HOME (SW_GAP is set to it in the Sensors steps). Turning on pulls the carriage back smoothly; at full cock the cut last tooth lets go (push the carriage home by hand). Hold the drive gear from the side and never put a hand in front of the carriage.

Step 50 (launcher 202). Launcher 9: tether pulleys

Press a 693ZZ into each printed pulley. Fit the top and bottom pulleys in the fork behind the rail (M3 x 20 axle and nyloc nut, a printed top-hat washer each side between plate and bearing, hub towards the bearing) and the pivot pulley on the right cheek lug under the axis (M3 x 20 from outside, head on the bearing, nyloc on the cheek inner face). Thread the line from the reel side under the pivot pulley, back under the rail to the bottom pulley, up over the top pulley and forward through the pusher hole to the dart's tail cap. Never fire without the tether.

Parts: launch_pulley_top, launch_pulley_top_brg, launch_pulley_top_axle, launch_pulley_bot, launch_pulley_bot_brg, launch_pulley_bot_axle, launch_pulley_piv, launch_pulley_piv_brg, launch_pulley_piv_axle, launch_pulley_top_washer_l, launch_pulley_top_washer_r, launch_pulley_bot_washer_l, launch_pulley_bot_washer_r, launch_line_b, launch_line_c, launch_line_p, launch_line_a0, launch_line_a1, launch_line_a2, launch_line_a3, launch_line_a4, launch_line_a5, launch_line_a6, launch_line_a7, launch_line_a8.

Step 51 (aim 205). Aim 1: aim housing, worm, wheel, crank

Build the aim housing on the bench (it goes onto the base later, after the rail). The AS5600 carrier and SW_RAIL_LOW are cad-sensors parts (Sensors steps), screwed into the pilots on the housing. Push the countershaft in from the inboard side through the inner wall bushing, through the big bevel (D bore, slid on from the outboard end of the shaft) and into the outer wall bushing. Drop the M8 nut into the hex pocket on the inner face of the outer wall, fit the two 608 bearings into the crank (worm wheel) hub, hold the crank in place and push the M8 x 25 bolt in from outboard through the wall, the nut and both bearings; tighten so the head and nut clamp the wall. Lay the worm shaft with the small bevel in its saddles and screw the cap on (4 x M3 x 30 self tap). The housing screws to the deck later (Aim 3), from the top: 4 x M3 x 10 down through the counterbored holes in its foot flanges (two behind the spine, one in front of the outer wall, one on the inboard tab) into the deck's heat-set inserts. The SW_RAIL_LOW bracket on the foot extension has its own deck screw (Sensors 2).

Parts: launch_aim_housing, launch_aim_cap, launch_aim_worm, launch_aim_crank, launch_aim_bolt, launch_aim_brg_1, launch_aim_brg_2, launch_aim_output, launch_aim_bevel30, launch_aim_nut.

Step 52 (aim 206). Aim 2: splitter

Slide the sun onto the sleeve's outer rib. On the bench: fit the idler on the swing arm pin and snap the printed C clip into the pin groove, and push the printed drag pad into the arm hub pocket with the shim behind its root block (a thicker shim gives more drag). Then slide the arm onto the sleeve next to the sun (there is no room to fit the idler once the arm is on).

Parts: launch_aim_sun, launch_aim_arm, launch_aim_idler, launch_aim_drag_pad, launch_aim_drag_shim, launch_aim_idler_clip.

Step 53 (aim 207). Aim 3: lug, conrod, rail onto the base

With the sun and swing arm already on the sleeve (the lug is on the rail since Launcher 2b), slide the rail assembly's sleeve along the main axis onto the drum's d12 spigot from the outboard side (the LAUNCH dog ring is already bolted to its flange, Launcher 4) until the ring's dogs sit by the drum collar. Hang the conrod's upper eye on the lug pin and snap a printed C clip into that groove. Then slide the aim housing on from the outboard side so the sleeve end enters its bushing and the crank pin enters the conrod's lower eye (hold the swing arm with the idler pointing down and a little to the rear, clear of the output gear, and let the arm go in between the two stop posts on the housing's inner face as they come up to it); snap a printed C clip onto the crank pin and screw the housing to the deck from the top: 4 x M3 x 10 through the counterbored foot flange holes into the deck inserts. Turn the sleeve backward by hand: the arm swings the idler into the output gear, comes to rest on its engaged stop post and drives the gear without binding; forward, the arm swings out onto the other post and the output gear stays still. Then turn the wiper backward slowly: the rail rises to 47 deg, then lowers to 6 deg.

Parts: launch_aim_conrod, launch_aim_crankpin_clip, launch_aim_lugpin_clip.

Step 54 (aim 208). Launcher 10: wiring (battery unplugged)

Battery unplugged. Run the SW_HOME, SW_SEATED and SW_GAP wires along the rail to the pivot and leave a 150 mm loop at the main axis before they go to the frame. No wire goes onto the carriage. Each contact: one wire to its ESP32 pin (SW_HOME 21, SW_GAP 39, SW_RAIL_LOW 40, SW_SEATED 41, as params.SWITCHES), the other to GND.

Step 55 (sensors 210). Sensors 1: build the two contact switches (SW_RAIL_LOW, SW_GAP)

Battery unplugged. For each switch: press an M3 nut into the hex trap on the base pillar (a drop of glue), solder a wire to the nut through the side slot and lay it in the groove. Screw the lever onto the pad with 2 x M3x10 self tapping screws. Screw an M3x10 through the lever boss from the top, head down, with a ring lug and the second wire under a jam nut on top. Turn the screw until the head touches the nut, then back it off 1.9 turns (gap about 0.9 mm, 0.5 mm per turn) and lock the jam nut.

Parts: sensors_sw_rail_low, sensors_sw_rail_low_lever, sensors_sw_rail_low_screw, sensors_sw_rail_low_jam, sensors_sw_rail_low_nut.

Step 56 (sensors 211). Sensors 2: SW_RAIL_LOW on the aim housing

Screw the switch base to its bracket (4 x M3x8 self tapping), lever pointing to the crank. Set the bracket foot on the housing foot extension in front of the crank. Screw it down with an M3x10 in the hole nearest the crank (self taps into the housing) and an M3x16 in the counterbored hole at the front end of the foot: down through the foot and the foot extension into the deck insert under it (the bracket's own deck screw). Turn the crank by hand (wiper backwards) to the lowest rail position: the cam lobe presses the lever tip and the meter shows the switch closed. Turn the crank about 20 deg either side: the switch must open.

Parts: sensors_sw_rail_low_bracket.

Step 57 (sensors 212). Sensors 3: SW_GAP on the right rail cheek

Build SW_GAP as in Sensors 1 (its base has its own arm). Hold the arm tab on the outer face of the right rail cheek, below the big gear, and screw it on with 2 x M3x12 (heads in the counterbores). With the big gear at rest (toothless gap at the rack) the cam bump presses the lever: switch closed. Turn the gear 20 deg either way: switch open.

Parts: sensors_sw_gap, sensors_sw_gap_lever, sensors_sw_gap_screw, sensors_sw_gap_jam, sensors_sw_gap_nut.

Step 58 (sensors 213). Sensors 4: AS5600 board

Battery unplugged. Do this after the magnet cap is on the main shaft end (drum steps): the board sits 2 mm in front of it. Battery unplugged. Plug the STEMMA QT cable into the board's connector on the +x end (the end away from the housing spine). Push the board onto the 4 pegs of the carrier, chip facing out. Drop an M3 nut into each hex pocket in the carrier back, lay the strips over the board edges (pegs through their holes) and screw each from the strip front with an M3x12 into its nut. Screw the carrier column to the aim housing spine with 2 x M3x16. Check by eye: the chip sits on the shaft axis, about 2.0 mm from the magnet face, and the cap turns free.

Parts: sensors_as5600, sensors_as5600_carrier, sensors_as5600_strip_top, sensors_as5600_strip_bot, sensors_as5600_nut_top, sensors_as5600_nut_bot, sensors_as5600_screw_top, sensors_as5600_screw_bot, sensors_as5600_col_screws.

Step 59 (electronics 220). Electronics bay

Battery unplugged. Bay on the deck, 4 x M3 x 10 through the tabs. Boards in: BTS7960 on its 4 standoffs (M3 x 6), heatsink down; ESP32 DevKitC pins down into the cradle, USB end into the wall slot, clamp with 2 x M3 x 8; DRV8833, MP1584 and HX711 into their pockets with a cable tie through the floor slots or a drop of hot glue. Fuse holder in its cradle with a cable tie.

Parts: frame_ebay_lid, frame_ebay_esp_clamp, frame_pcb_esp32, frame_pcb_bts7960, frame_pcb_mp1584, frame_pcb_drv8833, frame_pcb_hx711, frame_fuse_20a.
Also handle (fitted earlier): frame_ebay.

Step 60 (electronics 221). Set the buck BEFORE wiring the DRV8833

Battery unplugged. Wire only the buck input to the fuse output and ground. Plug the battery in, meter on the buck output, turn the trimpot until it reads 9.0 V. Unplug the battery. Only then wire buck output to DRV8833 VIN (the DRV8833 dies above 10.8 V).

Also handle (fitted earlier): frame_pcb_mp1584, frame_pcb_drv8833.

Step 61 (electronics 222). Camera, receiver and cable anchors

Battery unplugged. Camera column on the deck, 4 x M3 x 10 into the inserts. Clip the Kiyo X onto the 16 mm edge plate like onto a monitor (or use a 1/4-20 tripod screw up through the plate hole); it looks forward along the launch line. RC832: stand it antenna-end up in its holder slot, leads out of the low notch, holder 2 x M3 x 10. Cable anchors 1 x M3 x 12 each; tie the camera USB, the receiver AV lead and the capture dongle lead to them so no plug carries the cable weight.

Parts: frame_camera_mount, frame_rc832_holder, frame_rc832, frame_kiyo_x, frame_cable_anchor_1, frame_cable_anchor_2.

Step 62 (electronics 223). Wiring and lid

Battery unplugged. Follow firmware/base/WIRING.md. Every wire leaving the bay goes through its wall exit and is cable-tied to the anchor beside it (strain relief). 14 AWG for battery, fuse, BTS7960 and motor. Lid on, 4 x M3 x 6. USB cable from the ESP32 native USB port to the Jetson.

Also handle (fitted earlier): frame_ebay_lid.

Step 63 (guards 230). Gear guards

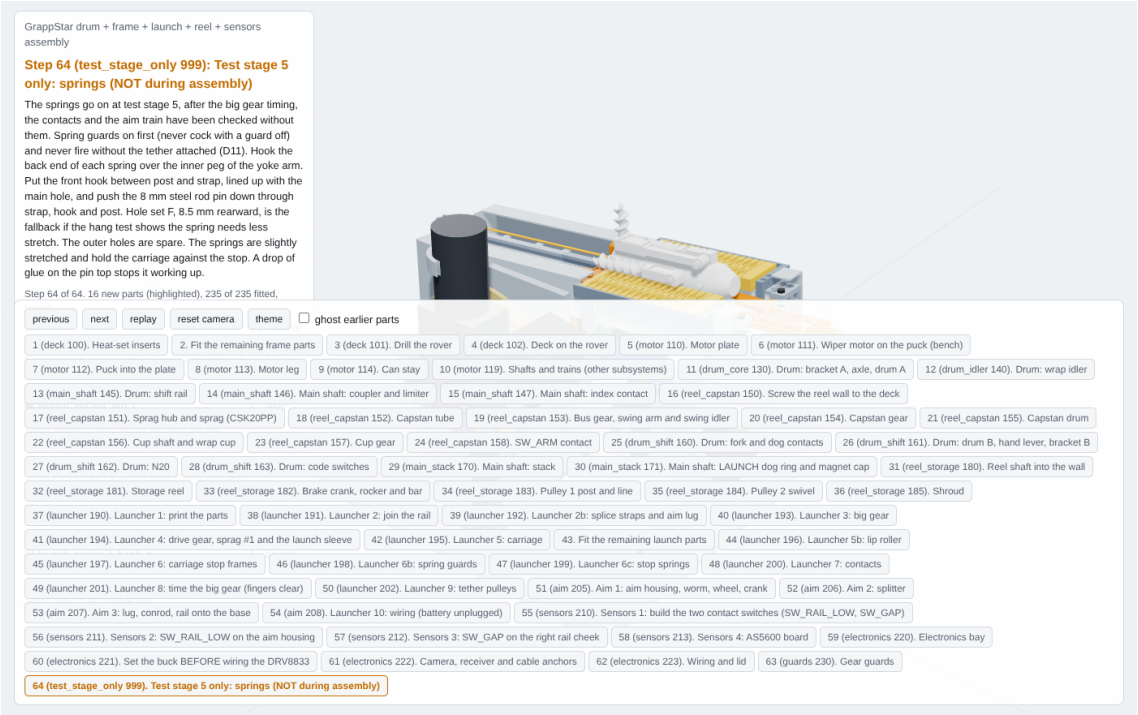
After every train has been turned by hand: the hood goes over the gears in front of the motor plate, M3 x 10 through each lug into the plate inserts; the front guard stands on the deck between the drum gears and the bay, M3 x 10 into the deck inserts. Turn the wiper by hand once: nothing rubs.

Parts: frame_guard_hood, frame_guard_front.

Step 64 (test_stage_only 999). Test stage 5 only: springs (NOT during assembly)

The springs go on at test stage 5, after the big gear timing, the contacts and the aim train have been checked without them. Spring guards on first (never cock with a guard off) and never fire without the tether attached (D11). Hook the back end of each spring over the inner peg of the yoke arm. Put the front hook between post and strap, lined up with the main hole, and push the 8 mm steel rod pin down through strap, hook and post. Hole set F, 8.5 mm rearward, is the fallback if the hang test shows the spring needs less stretch. The outer holes are spare. The springs are slightly stretched and hold the carriage against the stop. A drop of glue on the pin top stops it working up.

Parts: launch_anchor_pin_l, launch_anchor_pin_r, launch_spring_l_hook_a, launch_spring_r_hook_a, launch_spring_l_hook_c, launch_spring_r_hook_c, launch_spring_l_0, launch_spring_l_1, launch_spring_l_2, launch_spring_l_3, launch_spring_l_4, launch_spring_r_0, launch_spring_r_1, launch_spring_r_2, launch_spring_r_3, launch_spring_r_4.



6. Audit summary

Result: NOT CLEAN

235 parts checked over 44678 states.

Collision drum_coupler with drum_index_lever: 24.517 mm3 in 67 states.

Collision launch_aim_sun with launch_aim_idler: 55.565 mm3 in 12 states.

Allowed contacts: 97.